

PVC SUSPENSION REACTOR VER 1

Operator Training Simulator · batch suspension-PVC reactor R-101

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the R-101 batch PVC reactor

DOCUMENT NO. : PVC-V1-SA-HZP-001
DOCUMENT TITLE : HAZOP REPORT
STANDARD : IEC 61882:2016
DRAWINGS : PVC-PID-001 · PVC-PID-002
STUDY TYPE : Teaching study, basic-design level

Teaching document generated from the HAZOP tab of the R-101 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	06-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator · Document PVC-V1-SA-HZP-001 rev A · 39 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N5 Jacket tempered-water loop with CW and LP-steam injection	R01	N5.1, N5.2	Auto-start the spare jacket pump P-101B on P-101A trip or FI-110 low, completing well inside the FSLL-110 delay; test the change-over monthly.	Instrument / Operations		OPEN	Risk 10 High
N4/N8 R-101 during the reaction / Short-stop, emergency vent and relief	R02	N4.6, N5.1, N8.4	Put agitator M-101 and P-101A/B on the essential (emergency) power supply and prove by test that the short-stop is mixed after a power failure (P&ID note "agitator on emergency power" is still TBC).	Electrical		OPEN	Risk 10 High
N4 R-101 during the reaction	R03	N4.6, N4.7	Detect loss of agitation independently of the motor: impeller shaft-speed switch, with "low power while running" (JI-100) taken to LOPA as a candidate SIS initiator.	Instrument		OPEN	Risk 10 High
N3 Initiator charge	R04	N3.1, N3.7	Independent check of the initiator charge: weigh D-104 before and after dosing with a high-mass interlock on XV-103, and latch the dose so it can be given only once per batch.	Instrument / Process		OPEN	Risk 10 High
N3 Initiator charge	R05	N3.6	Organic peroxide storage: refrigerated store with independent high-temperature alarm and back-up cooling, inventory limited to one shift at D-104, supplier SADT and emergency temperatures posted, separation from the reactor bay.	Process / Safety		OPEN	Risk 8 Medium
N4/N5/N8 R-101 during the reaction / Jacket tempered-water loop with CW and LP-steam injection / Short-stop, emergency vent and relief	R06	N3.1, N4.1, N5.7, N7.1, N8.1	LOPA for the runaway scenarios that rely on TSHH-101, PSHH-101, XS-101, FSLL-110 and the short-stop; set the SIL target and the proof-test interval of XV-104 and the logic solver.	Process safety		OPEN	Risk 10 High
N8 Short-stop, emergency vent and relief	R07	N8.1, N8.2	Add a pressure transmitter with low-pressure alarm on the short-stop pot D-105 and make it an initiation permissive.	Instrument		OPEN	Risk 5 Medium
N8 Short-stop, emergency vent and relief	R08	N8.6, N8.7	Confirm PSV-100 / XV-108 emergency-relief sizing for the runaway case with two-phase flow (DIERS method) and the capacity of D-103 for one full relief.	Process		OPEN	Risk 5 Medium
N5/N6 Jacket tempered-water loop with CW and LP-steam injection / Baffle cooling water FIC-120	R09	N5.7, N5.8, N6.1	CW supply: low header-pressure alarm and CW supply flow on the reactor DCS; study CW failure as a common cause for all four reactors (stagger peak heat release, back-up CW pump on essential power).	Process / Utilities		OPEN	Risk 10 High
N4 R-101 during the reaction	R10	N4.1, N4.3	Deviation alarm TE-100 vs TE-101 (1 °C) so a drifting control element is seen before the K-value or the trip margin is lost.	Instrument		OPEN	Risk 10 High
N4/N10 R-101 during the reaction / Procedures	R11	N4.5, N9.4, N10.5	Deviation alarm PT-100 vs PT-101 and a frozen-value check on PI-100, because the PD-100 end point depends on PT-100 alone.	Instrument		OPEN	Risk 8 Medium
N7/N8 Reflux condenser E-101 and inert vent / Short-stop, emergency vent and relief	R12	N4.4, N7.4, N8.3	Double block and bleed on the nitrogen supply to D-105 (open only to pressurise the pot); consider automatic inert venting on TI-132 low with a flow limit on HIC-135.	Process / Instrument		OPEN	Risk 8 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N7 Reflux condenser E-101 and inert vent	R13	N7.2	Calculated E-101 duty check (FI-130 × ΔT against the heat balance) with a deviation alarm; include FI-130 in the calibration plan.	Instrument		OPEN	Risk 9 Medium
N7 Reflux condenser E-101 and inert vent	R14	N7.1, N7.2	Car-seal open the E-101 CW block valves and add FI-130 LO alarm while E-101 is in service.	Operations / Instrument		OPEN	Risk 10 High
N4/N6/N7 R-101 during the reaction / Baffle cooling water FIC-120 / Reflux condenser E-101 and inert vent	R15	N4.10, N4.11, N6.2, N6.5, N7.3, N7.5	Fouling programme: wall, baffle and E-101 cleaning interval set from the TV-110B and TI-131 trends; keep the VPC-130 anti-foam limit in service.	Operations		OPEN	Risk 9 Medium
N1/N2/N4 Demineralised water and PVA charge / VCM charge / R-101 during the reaction	R16	N1.2, N2.2, N4.8, N4.9	Confirm LSHH-100 uses its own level sensor (not LT-100) and add a check of LIC-100 against the load cells WI-100.	Instrument		OPEN	Risk 8 Medium
N4 R-101 during the reaction	R17	N4.8	Second shut-off valve on the injection-water line closing on LSHH-100, and an alarm "FI-103 flow with FIC-103 output at zero".	Instrument		OPEN	Risk 8 Medium
N5 Jacket tempered-water loop with CW and LP-steam injection	R18	N5.4, N5.6	Tight shut-off block valve on LP steam to SM-101, closed while the batch reacts, with position indication on the DCS.	Process / Instrument		OPEN	Risk 9 Medium
N5 Jacket tempered-water loop with CW and LP-steam injection	R19	N5.4	Valve-position feedback on TV-110B with a deviation alarm against the TIC-110 output (stuck valve detection).	Instrument		OPEN	Risk 8 Medium
N1/N2/N10 Demineralised water and PVA charge / VCM charge / Procedures	R20	N1.7, N2.4, N10.1	Vacuum-hold test (pressure rise over 5 min) before the VCM charge; oxygen check of recovered VCM before re-use.	Operations		OPEN	Risk 8 Medium
N2 VCM charge	R21	N2.6, N4.12	Remotely operated isolation valve at the TK-100 outlet and VCM gas detectors at P-103, the charge floor and the agitator seal, with automatic isolation of the charge.	Process / Instrument		OPEN	Risk 10 High
N9 Degassing to VCM recovery and discharge to D-102	R22	N9.3	Make short-stop (or end point reached) a hard permissive for opening XV-105; today the page only gives a warning alarm.	Instrument		OPEN	Risk 6 Medium
N9 Degassing to VCM recovery and discharge to D-102	R23	N9.4, N9.5, N9.7	Route the D-102 vent to VCM recovery, add D-102 high-level alarm and confirm D-102 holds one full batch plus margin.	Process		OPEN	Risk 8 Medium
N10 Procedures	R24	N9.9, N10.7, N10.8	Reactor opening: degas to recovery, steam-strip and evacuate until a portable VCM measurement at the manway is below the occupational limit; closed-reactor high-pressure water cleaning so that man-entry is the exception; confined-space permit with positive isolation.	Operations / HSE		OPEN	Risk 12 High
N1/N2/N10 Demineralised water and PVA charge / VCM charge / Procedures	R25	N1.2, N1.3, N1.6, N2.2, N10.9	Recipe presets (FQI-101, FQI-102, initiator) downloaded from a recipe system and checked by a second person before the charge.	Operations		OPEN	Risk 9 Medium
N7/N9/N10 Reflux condenser E-101 and inert vent / Degassing to VCM recovery and discharge to D-102 / Procedures	R26	N7.6, N7.7, N9.2, N9.8, N10.2	Show VCM recovery status (compressor C-101 running, gasholder D-106 level) on the reactor DCS; keep vacuum-pump exhaust away from the recovery system (vent treatment only).	Process		OPEN	Risk 8 Medium
All whole unit	R27	N3.1, N3.5, N4.7, N5.9, N7.9, N10.4	Train operators on the OTS for loss of agitation, loss of cooling, initiator overcharge and hot initiation, including when to short-stop by hand.	Operations		OPEN	Risk 10 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N8 Short-stop, emergency vent and relief	R28	N2.4, N8.9	Trip-bypass management: any SIS bypass needs a permit, a time limit and MOC; bypass status shown on the DCS.	Operations		OPEN	Risk 10 High
N1 Demineralised water and PVA charge	R29	N1.6	Measure the PVA charge (flow totaliser or TK-107 load cell) and make it a VCM-charge permissive.	Instrument		OPEN	Risk 9 Medium
N1/N2/N3 Demineralised water and PVA charge / VCM charge / Initiator charge	R30	N1.9, N2.7, N3.8	Check valves plus the FC charge valves on every line into R-101 (water, VCM, initiator); confirm back-flow protection because R-101 runs above the water header and the storage pressure.	Process		OPEN	Risk 8 Medium
N2 VCM charge	R31	N2.5, N2.6	Minimum-flow recycle to TK-100 on P-103 and a double mechanical seal with barrier-pressure alarm.	Mechanical		OPEN	Risk 10 High
N2 VCM charge	R32	N2.8	Specification check of recovered VCM before re-use (oxygen, acetylene, HCl, water).	Laboratory		OPEN	Risk 6 Medium
N3 Initiator charge	R33	N3.2, N3.6, N3.7	Initiator lot control: assay certificate, first-in-first-out, store temperature records; reject drums that have been warm.	Laboratory / Operations		OPEN	Risk 8 Medium
N6/N7 Baffle cooling water FIC-120 / Reflux condenser E-101 and inert vent	R34	N6.4, N7.8	VCM detection on the CW returns from the baffles, jacket overflow and E-101 (tube-leak detection) before the cooling tower.	Instrument		OPEN	Risk 8 Medium
N8 Short-stop, emergency vent and relief	R35	N8.7	Rupture disc with pressure-monitored interspace under PSV-100 to stop polymer plugging the relief path; inspect the relief line by batch count.	Mechanical		OPEN	Risk 5 Medium
N8 Short-stop, emergency vent and relief	R36	N8.5	Review the spurious-trip rate of the SIS initiators (e.g. 2oo3 voting on temperature and pressure) together with the SIL target from R06.	Instrument		OPEN	Risk 6 Medium
N9 Degassing to VCM recovery and discharge to D-102	R37	N9.1	Degassing rate limit (ramp on HIC-105) and a knock-out pot ahead of the gasholder to catch foam and slurry.	Process		OPEN	Risk 9 Medium
N10 Procedures	R38	N8.4, N10.10	Written rule for restarting the agitator after it has stopped on a batch that is still reacting (only after the short-stop has been confirmed, with full cooling), practised on the OTS.	Operations		OPEN	Risk 10 High
N8 Short-stop, emergency vent and relief	R39	N8.8	Dedicated instrument-air receiver (with check valve) for XV-104 and XV-108 so the short-stop and emergency vent can still open after an air failure.	Instrument		OPEN	Risk 6 Medium

HAZOP REPORT

Hazard and Operability Study of the R-101 batch suspension-PVC reactor
PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the R-101 reactor drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

CONTENTS

HAZOP CLOSE-OUT SUMMARY	2
Executive summary	8
1 Introduction	10
1.1 Purpose	10
1.2 The unit	10
1.3 Objectives	10
2 Scope	10
2.1 Nodes	10
2.2 Node map	11
3 Methodology	13
3.1 References	13
3.2 The four phases	13
3.3 Guide words and parameters	13
3.4 Risk ranking	14
4 HAZOP sessions	15
4.1 Schedule	15
4.2 Team	15
4.3 Documents reviewed	15
4.4 Nodes reviewed and guide-word coverage	15
5 Findings and recommendations	17
5.1 Risk profile	17
5.2 Recommendations	17
5.3 Dynamic check of selected deviations on the simulator	19
5.4 Follow-up and close-out	20
Appendix 1 · HAZOP worksheets	21
Node N1 · Demineralised water and PVA charge	21
Node N2 · VCM charge	23
Node N3 · Initiator charge	24
Node N4 · R-101 during the reaction	26
Node N5 · Jacket tempered-water loop with CW and LP-steam injection	28
Node N6 · Baffle cooling water FIC-120	30
Node N7 · Reflux condenser E-101 and inert vent	31
Node N8 · Short-stop, emergency vent and relief	33
Node N9 · Degassing to VCM recovery and discharge to D-102	34
Node N10 · Procedures	35
Appendix 2 · HAZOP documents (drawings with node boundaries)	37
A2-1 · HAZOP node map	37
A2-2 · Simulator P&ID (reference drawing of the DCS console and the exercises)	38
A2-3 · PVC-PID-001 (R-101, charges, condenser and vents)	39

A2-4 · PVC-PID-002 (heat removal: jacket loop, baffles, E-101 cooling water)	40
Appendix 3 · HAZOP action close-out sheets	41

Executive summary

A HAZOP of the R-101 batch suspension-PVC reactor — the water/PVA, VCM and initiator charges, the reactor during the reaction, the jacket tempered-water loop with its CW and steam injection, the baffle cooling, the reflux condenser E-101 and its inert vent, the short-stop, emergency vent and relief, degassing and discharge, and the batch procedures (evacuation, charging, initiation, end point, opening the reactor) — was carried out with the IEC 61882 guide-word method. The unit was divided into **10 nodes**; **91 deviations** were recorded and **39 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
10	91	39	0	12	61	18

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the high-ranked deviations are those where the short-stop or the relief valve is the main protection against a runaway or a liquid-full reactor, and they are carried to the LOPA, the SRS of the SIFs (PVC-V1-IN-SRS-001) and the cause and effect diagram (PVC-V1-IN-CE-001).

Key findings

- **Agitator stops (N4):** XS-101 short-stops the batch at 00:15; the trip is the first annunciation (JI-100 LO is suppressed once the motor is off). TI-101 then falls to 45.5 °C and PI-101 to 6.30 bar g. Without the impeller the short-stop mixes in minutes rather than seconds; with full cooling the batch still cools here because the stop came at 2 h ($X \approx 26\%$), well before the heat-release peak — near the gel peak the margin would be much smaller, hence R02 (essential power) and N8.4.
- **Agitation lost with the motor running (N4):** JI-100 LO at 00:01 is the only alarm and nothing trips. The controllers hold TI-101 at 57.0–57.2 °C, so temperature does not warn anyone, while the coarse fraction reaches 16.0 % in 30 min (R03).
- **Jacket pump trips (N5):** FI-110 LO at 00:02, FSLL-110 short-stops the batch at 00:36: about 34 s for the operator to start P-101B (R01).
- **Cooling water lost (N5):** FIC-120 LO at 00:01, TIC-100 HI at 06:02, TI-101 HI at 13:23, TSHH-101 at 20:45 with PI-101 at 10.10 bar g. Roughly 21 min from the loss to the trip — time for a controlled short-stop by hand, but on all four reactors at once (R09).
- **Double initiator dose (N3):** heat release climbs to 9.81 MW against 2.71 MW at design; TIC-100 HI at 04:38, TI-101 HI at 11:34, TSHH-101 at 17:30. The PSV does not lift: the trip protects, but only an independent dose check prevents it (R04).
- **E-101 cooling water isolated (N7):** TV-110B reaches 100 % at 02:48, TIC-100 HI at 06:22 and TSHH-101 trips at 25:30. At 2 h the jacket alone cannot hold the batch, so E-101 cooling is safety-relevant: N7.1 is ranked High and R14 (car-seal, FI-130 LO) follows.
- **Nitrogen passing into R-101 (N7):** the only alarm is TI-132 LO at 24:18; VPC-130 quietly raises the E-101 water from 36 to 69 m³/h and PI-101 rises to 9.62 bar g. The controller hides the loss of condenser area (R12).
- **TE-100 drifting low (N4):** TIC-100 never alarms (its own PV stays near 57 °C), while the real temperature reaches 60.2 °C; the first warnings are TI-131 HI at 10:05 and TI-101 HI at 13:05. Only the independent TE-101 shows it — hence the deviation alarm R10.
- **Jacket CW valve driven fully open (N5):** TIC-100 LO at 05:10; the batch cools to 49.8 °C and the heat release falls to 1.45 MW — the safe direction, but off grade and slower.
- **Instrument air lost (N8):** no trip; the fail-open cooling valves give full cooling (FI-130 up to 540 m³/h) and the batch cools to 48.6 °C. The fail positions are right for cooling, but XV-104 and XV-108 (fail closed) could not open if a runaway came from another cause at the same time (R39).

Actions by action party

Action party	Number	Action No.
Instrument	19	R01, R03, R04, R07, R10, R11, R12, R13, R14, R16, R17, R18, R19, R21, R22, R29, R34, R36, R39
Process	11	R04, R05, R08, R09, R12, R18, R21, R23, R26, R30, R37
Operations	10	R01, R14, R15, R20, R24, R25, R27, R28, R33, R38
Mechanical	2	R31, R35
Laboratory	2	R32, R33
Electrical	1	R02

Action party	Number	Action No.
Safety	1	R05
Process safety	1	R06
Utilities	1	R09
HSE	1	R24

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how R-101 can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that harm no one but lose a batch, put the resin off grade (K-value, lumps) or upset the VCM recovery. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram PVC-V1-IN-CE-001.

1.2 The unit

R-101 is one of four 140 m³ batch reactors. Each batch charges 60 t of demineralised water with the PVA suspending agent, evacuates the reactor, charges 51 t of vinyl chloride monomer (VCM), heats to 57 °C and doses the peroxide initiator. The reaction runs about 5.3 h at 57 °C and 8.6 bar g (the VCM vapour pressure) until the end-point detector PD-100 sees the pressure fall 1.8 bar; the batch is then short-stopped, degassed to VCM recovery and discharged to D-102. Heat is removed by the jacket tempered-water loop (P-101A/B, CW injection TV-110B and steam TV-110A on split range under the TIC-100 → TIC-110 cascade), the baffles (FIC-120) and the reflux condenser E-101 (FIC-130 under the valve-position controller VPC-130); heat release peaks at about 3.4 MW near 72 % conversion. The SIS short-stops the batch through XV-104 from pot D-105 on high temperature (TSHH-101), high pressure (PSHH-101), loss of agitation (XS-101) or loss of jacket circulation (FSL-110), and shuts the charges on high-high level (LSHH-100); PSV-100 relieves to the dump tank D-103.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node and each batch step.
- Identify operability problems: lost batches, off-grade resin, lumps and wall build-up, upsets of the VCM recovery.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope R-101 from the charge valves (water, VCM, initiator, short-stop) to XV-105 (recovery header) and XV-106 (to D-102), including the jacket loop, baffles, E-101 and its vent, the SIS and relief, and the batch procedures; excluded: TK-100 store internals, the recovery plant, D-102 onwards, the other three reactors (covered by the HAZID). Objectives, team, risk matrix.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Demineralised water and PVA charge	Charge 60 t of demineralised water at 55 °C through FIC-101 (FV-101 fail closed) and XV-101 (fail closed) until FQI-101 reaches its preset, with the PVA suspending agent from pot TK-107 (HV-107) into the evacuated R-101.	60 t at 55 °C, FIC-101 set point 240 m ³ /h, water header 5.0 bar g; R-101 under vacuum	Demineralised-water header, TK-107 PVA pot, HV-107, FV-101, XV-101, charge line to R-101 <i>FIC-101, FQI-101 (preset), LIC-100, PI-100, WI-100</i>
N2	VCM charge: TK-100 → P-103 → FV-102 / XV-102 → R-101	Charge 51 t of VCM at 25 °C from storage TK-100 with P-103 through FIC-102 (FV-102 fail closed) and XV-102 (fail closed) until FQI-102 reaches its preset; only with the reactor evacuated, water in, M-101 running and the jacket circulating.	51 t at 25 °C, FIC-102 set point 180 t/h, pump discharge 12.0 bar g; level after charge ≈ 83 %	TK-100, P-103, FV-102, XV-102, VCM charge line <i>FIC-102, FQI-102 (preset), PI-100, LIC-100, LSHH-100</i>
N3	Initiator charge: D-104 → P-104 → XV-103	With the VCM charged and TI-100 ≥ 54 °C, dose the peroxide initiator mix (fast A, half-life 1.5 h, and slow B, 12 h, at 57 °C) from the chilled pot D-104 with P-104 through XV-103 in 60 s; the reaction time starts.	One dose per batch in 60 s; D-104 kept chilled; initiation permissive TI-100 ≥ 54 °C and FQI-102 ≥ 95 %	D-104 (chilled), P-104 dosing pump, XV-103, initiator line; peroxide store <i>Batch sequence (initiator permissive), TI-100, XI-100</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N4	R-101 during the reaction (temperature, pressure, level, agitation, composition)	Hold 57 °C (grade K 67, band ± 1 °C) at about 8.6 bar g with the level at 83 % (LIC-100 → FIC-103 shrinkage water) and M-101 running, for ≈ 5.3 h until PD-100 sees a 1.8 bar drop below the running pressure; heat release peaks at ≈ 3.4 MW near 72 % conversion.	57.0 °C, 8.62 bar g, level 82.9 %, M-101 290 kW, heat release 2.71 MW at 2.0 h	R-101 (140 m ³ , design 15.2 bar g / 93 °C), agitator M-101 (330 kW), FV-103 injection <i>TIC-100 (TE-100), TI-101 / TSHH-101 (TE-101), PI-100 (PT-100), PI-101 / PSHH-101 (PT-101), LIC-100, LSHH-100, WI-100, JI-100, XS-101, XI-100, PD-100, KX-100</i>
N5	Jacket tempered-water loop with CW and LP-steam injection	Circulate tempered water (P-101A running, P-101B spare) through the jacket; TIC-110 (slave of TIC-100) sets the jacket inlet by split range: 0-50 % TV-110A LP steam (fail closed) into mixer SM-101, 50-100 % TV-110B CW injection (fail open); the surplus overflows to CW return.	FI-110 473 m ³ /h, jacket in 38.2 °C / out 40.9 °C, TV-110B 53 % open, jacket duty 1.50 MW	P-101A/B, SM-101, TV-110A, TV-110B, jacket (5.2 bar g), loop piping <i>TIC-110 (TE-110), TI-111, FI-110 / FSLL-110, TI-120</i>
N6	Baffle cooling water FIC-120	Cool the four internal baffles with CW through FIC-120 (FV-120 fail open) at 80 m ³ /h; return monitored by TI-121.	FIC-120 80 m ³ /h, return 36.3 °C, baffle duty 0.40 MW	FV-120, four baffles, CW supply and return <i>FIC-120, TI-121, TI-120</i>
N7	Reflux condenser E-101 and inert vent	Condense VCM boiling off the slurry in E-101 on CW (FIC-130, FV-130 fail open) with the condensate running back; VPC-130 moves the E-101 water to hold the jacket CW valve TV-110B near 50 % (anti-foam limit 12 m ³ /h per % XI-100); inerts vented by HIC-135 (fail closed) to VCM recovery at 1.25 bar a.	E-101 duty 0.97 MW, FIC-130 36 m ³ /h, TI-131 55.0 °C, TI-132 54.0 °C	E-101 (vertical, VCM in tubes), FV-130, HV-135, vapour line, vent to recovery <i>FIC-130 (FT-130), VPC-130, TI-131, TI-132, HIC-135</i>
N8	Short-stop, emergency vent and relief	On demand (operator, or SIS on TSHH-101 / PSHH-101 / XS-101 / FSLL-110) open XV-104 to inject the short-stop from the N ₂ -pressurised pot D-105 into R-101, where the agitator mixes it within a minute; ESD also opens the emergency vent XV-108; PSV-100 at 15 bar g protects the vessel.	Trips: 63 °C, 12.5 bar g, agitator stopped 15 s, circulation < 120 m ³ /h 30 s; PSV 15 bar g; vessel 15.2 bar g / 93 °C	D-105, XV-104, XV-108, PSV-100, dump / emergency vent tank D-103, SIS logic solver <i>TSHH-101, PSHH-101, XS-101, FSLL-110, LSHH-100, HS-001 ESD</i>
N9	Degassing to VCM recovery and discharge to D-102	After the short-stop, degas unreacted VCM through XV-105 and HIC-105 / PV-105 (fail closed) to recovery (gasholder D-106 at 1.25 bar a, compressor C-101) to below 2 bar g, pad with N ₂ through HV-141 / PCV-141 (1.5 bar g), then discharge the slurry through XV-106 (fail closed) to blowdown tank D-102 with the agitator running.	Recovery at 1.25 bar a; discharge permitted below 2.0 bar g; N ₂ pad 1.5 bar g; batch ≈ 118 t	XV-105, PV-105, recovery header, D-106, C-101, HV-141/PCV-141, XV-106, D-102, P-102 <i>HIC-105, FI-105, PI-100, PD-100, WI-100, FI-140</i>
N10	Procedures: evacuation, charging, initiation, end point, opening the reactor	Batch sequence: evacuate with VP-101 (XV-109) → water + PVA → agitator → VCM → heat up on steam → initiator at ≥ 54 °C → reaction on cascade → end point → short-stop → degas → discharge; between batches clean and, when needed, open the reactor.	Procedures in the Exercises tab (cold start, end of batch); reactor opening by permit	Batch sequence, VP-101/XV-109, operator actions, reactor manway <i>Batch & SIS display, PI-100, PD-100, FQI-101/102, TIC-100, TIC-110</i>

2.2 Node map

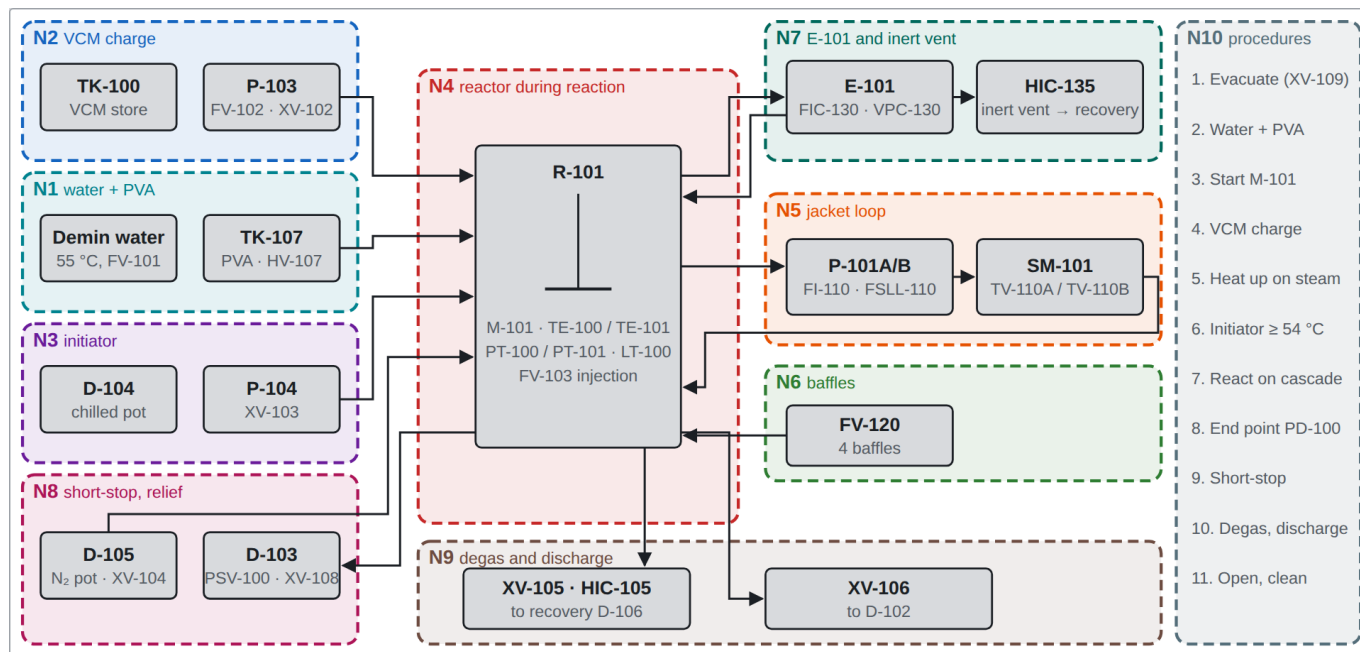


Figure 1. R-101 divided into HAZOP nodes (N10, the procedures node, covers the whole unit).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds both hazards (harm to people, environment, assets) and operability problems (the batch cannot be made to grade or on time). For a batch reactor the intent changes with the phase, so the time-related guide words (EARLY, LATE, BEFORE, AFTER) and a procedure node matter as much as NO flow or MORE temperature.

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i> (3rd ed.)	Worksheet format, node selection, batch and procedural HAZOP, "consequence without safeguards", risk ranking, links to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the short-stop trips TSHH-101, PSHH-101, XS-101 and FSLL-110
CCPS <i>Guidelines for Process Safety in Batch Reaction Systems</i> ; DIERS methodology	Runaway scenarios, short-stop systems, emergency relief sizing with two-phase flow
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable; occupational VCM standards (e.g. OSHA 1910.1017)	Process hazard analysis for covered processes, revalidated at least every five years; VCM exposure control at opening and cleaning

3.2 The four phases

- **Definition:** scope R-101 from the charge valves (water, VCM, initiator, short-stop) to XV-105 (recovery header) and XV-106 (to D-102), including the jacket loop, baffles, E-101 and its vent, the SIS and relief, and the batch procedures; excluded: TK-100 store internals, the recovery plant, D-102 onwards, the other three reactors (covered by the HAZID). Objectives, team, risk matrix.
- **Preparation:** collect the P&ID, PFD, design data, cause-and-effect (trip) chart, batch sequence and permissives, operating procedures, alarm list, datasheets; divide the plant into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** for each node → state the design intent → choose a parameter → apply each guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Record every meaningful deviation, including those with adequate safeguards ("full recording"), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, action close-out, and revalidation after changes (MOC) or periodically.

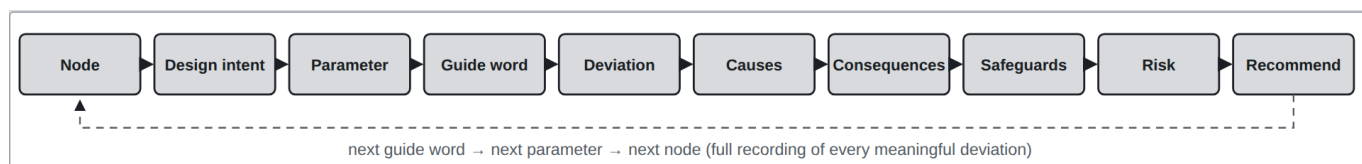


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)
NO / NOT	Complete negation of the intent
MORE	Quantitative increase
LESS	Quantitative decrease
AS WELL AS	Qualitative increase: something extra present
PART OF	Qualitative decrease: only part of the intent achieved
REVERSE	Logical opposite of the intent
OTHER THAN	Complete substitution: something else happens

Guide word	Meaning (IEC 61882)
EARLY	Relative to clock time
LATE	Relative to clock time
BEFORE	Relative to order or sequence
AFTER	Relative to order or sequence

Parameters used: flow, pressure, temperature, level, composition, agitation, heat transfer, measurement, control, utility, integrity (containment), safety system, and for procedures: sequence, timing, evacuation, charging, initiation, end point, opening, recipe, restart. A deviation is recorded only if it is physically meaningful ("REVERSE temperature" is not).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5 (6)	10 (11)	15	20	25
S4	4 (1)	8 (19)	12 (1)	16	20
S3	3 (3)	6 (13)	9 (11)	12	15
S2	2	4 (9)	6 (12)	8	10
S1	1	2	3 (5)	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk = S × L (the same matrix ranks the HAZID register).

S	Severity (worst credible consequence, no credit for safeguards)
1	No injury; batch delayed
2	First aid; off-grade batch; small contained release
3	Lost-time injury; reactor out for days (lumps, fouling); local release
4	Permanent disability or single fatality; VCM release with fire; serious occupational exposure
5	Multiple fatalities; runaway with vessel rupture or vapour-cloud explosion; escalation to the other reactors

L	Likelihood with the existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4} - 10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3} - 10^{-2} /y)
4	Likely: several times in the plant life (10^{-2} - 10^{-1} /y)
5	Frequent: yearly or more

1-4 Low acceptable · 5-9 Medium reduce if practicable (ALARP), review safeguards · 10-16 High recommendation required, consider LOPA · 20-25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N3 (water and PVA, VCM and initiator charges)		
2	N4 – N7 (R-101 during the reaction, jacket loop, baffles, E-101 and inert vent)		
3	N8 – N10 (short-stop and relief, degassing and discharge, procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, recipe, heat balance, cooling capacity, relief basis		
Operations representative	How batches are really run: charging, heat-up, end point, degassing, opening and cleaning		
Instrument / control engineer	Loops, fail positions, batch sequence and permissives, SIS logic and trip independence		
Safety / risk engineer	Risk matrix, LOPA follow-up, regulations, occupational VCM exposure		
Specialists as needed	Polymerisation chemist (initiator, short-stop, gel effect), mechanical (agitator, seals), initiator supplier (SADT)		

4.3 Documents reviewed

Document	Title / content
PVC-PID-001 rev A	R-101 P&ID sheet 1: reactor, charges, condenser and vents (Appendix 2)
PVC-PID-002 rev A	R-101 P&ID sheet 2: heat removal, jacket loop, baffles, E-101 cooling water (Appendix 2)
Simulator P&ID	The P&ID tab drawing used by the DCS console and the exercises (Appendix 2)
PFD-100	Process flow diagram with the stream flags (HAZID and PFD tabs)
PVC-V1-IN-CE-001 rev R0	Cause and effect diagram of the SIS, alarms and batch permissives
PVC-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design of the unit operation, Control loop design, alarm and trip list (Workbook tab), batch permissives (Batch & SIS tab), cold-start and end-of-batch procedures, SOP tab

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Parameters
N1	1	2	2	1	1	1	1	·	·	·	·	9	Flow, Temperature, Composition
N2	1	2	1	1	·	1	1	·	·	1	·	8	Flow, Sequence, Pressure, Integrity, Composition
N3	1	2	1	·	·	1	1	1	1	·	·	8	Flow, Timing, Temperature, Composition
N4	1	3	4	1	·	·	3	·	·	·	·	12	Temperature, Measurement, Pressure, Agitation, Level, Composition, Heat transfer, Integrity
N5	2	2	2	1	·	1	3	·	·	·	·	11	Flow, Temperature, Utility, Control, Measurement, Pressure

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Parameters
N6	1	2	2	.	.	.	1	.	.	.	.	6	Flow, Integrity, Heat transfer, Temperature
N7	1	3	2	1	.	.	2	.	.	.	.	9	Flow, Composition, Heat transfer, Pressure, Integrity, Control
N8	3	2	.	1	1	.	1	.	1	.	.	9	Flow, Timing, Pressure, Utility, Safety system
N9	1	2	1	.	.	1	1	1	.	1	1	9	Flow, Timing, Sequence, Agitation, Level
N10	.	.	1	.	.	.	4	2	1	1	1	10	Evacuation, Charging, Initiation, End point, Opening, Recipe, Restart

A dot means that guide word gave no meaningful deviation for any parameter in the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the charge nodes, the end of batch (N9) and the procedure node N10.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Demineralised water and PVA charge	9	0	5	4	R16, R20, R25, R27, R29, R30
N2 VCM charge	8	1	5	2	R16, R20, R21, R25, R27, R30, R31, R32
N3 Initiator charge	8	1	4	3	R04, R05, R27, R30, R33
N4 R-101 during the reaction	12	4	8	0	R02, R03, R06, R10, R11, R15, R16, R17, R27
N5 Jacket tempered-water loop with CW and LP-steam injection	11	2	6	3	R01, R06, R09, R18, R19, R27
N6 Baffle cooling water FIC-120	6	0	4	2	R09, R15, R27, R34
N7 Reflux condenser E-101 and inert vent	9	1	8	0	R12, R13, R14, R15, R26, R27, R34
N8 Short-stop, emergency vent and relief	9	2	7	0	R02, R06, R07, R08, R12, R27, R28, R35, R36, R39
N9 Degassing to VCM recovery and discharge to D-102	9	0	7	2	R22, R23, R26, R27, R37
N10 Procedures	10	1	7	2	R11, R20, R24, R25, R26, R27, R38
Total	91	12	61	18	39

High-ranked deviations

Ref	Deviation	Consequence	SxL	Rec.
N2.6	Integrity OTHER THAN: VCM leak from the charge system	Flashing VCM release: vapour cloud, flash fire or explosion; exposure to a carcinogen	5x2 = 10	R21, R31
N3.1	Flow MORE: More initiator than the recipe	Heat release beyond the cooling capacity: temperature and pressure rise, runaway, relief of VCM	5x2 = 10	R04, R06, R27
N4.1	Temperature MORE: Reactor hotter than 57 °C	K-value off grade, pressure rises with the VCM vapour pressure, runaway to relief	5x2 = 10	R06, R10
N4.6	Agitation NO: Agitator stops	Suspension separates within a minute: VCM droplets coalesce, jacket heat transfer collapses, hot spots and lumps; runaway without short-stop	5x2 = 10	R02, R03
N4.7	Agitation LESS: Agitation lost with the motor running	As above but XS-101 does not see it (motor shows RUN); lumps grow; heat transfer falls	5x2 = 10	R03, R27
N4.12	Integrity OTHER THAN: VCM leak from R-101	Flashing VCM release at 57 °C / 8.6 bar g: vapour cloud, flash fire or explosion; exposure to a carcinogen	5x2 = 10	R21
N5.1	Flow NO: No jacket circulation	Jacket cooling lost: temperature and pressure rise, runaway	5x2 = 10	R01, R02
N5.7	Utility NO: Cooling-water supply lost	Jacket, baffles and E-101 all lose cooling; runaway on every reacting batch	5x2 = 10	R09, R06
N7.1	Flow NO: No cooling water to E-101	E-101 duty lost; the jacket alone cannot hold the peak: temperature and pressure rise to the trip	5x2 = 10	R14, R06
N8.4	Timing LATE: Short-stop injected when the agitator has already stopped	Short-stop does not mix (minutes instead of seconds): local reaction goes on in the separated mass	5x2 = 10	R02, R38
N8.9	Safety system OTHER THAN: SIS initiator bypassed	The runaway protection is missing when needed	5x2 = 10	R28
N10.7	Opening AFTER: Reactor opened before VCM is removed	Operators exposed to VCM, a carcinogen with a very low occupational limit; flammable atmosphere at the manway	4x3 = 12	R24

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	Auto-start the spare jacket pump P-101B on P-101A trip or FI-110 low, completing well inside the FSLL-110 delay; test the change-over monthly.	N5	N5.1, N5.2	Instrument / Operations
R02	Put agitator M-101 and P-101A/B on the essential (emergency) power supply and prove by test that the short-stop is mixed after a power failure (P&ID note "agitator on emergency power" is still TBC).	N4/N8	N4.6, N5.1, N8.4	Electrical

No.	Recommendation	Node	Raised from	Action by
R03	Detect loss of agitation independently of the motor: impeller shaft-speed switch, with "low power while running" (JI-100) taken to LOPA as a candidate SIS initiator.	N4	N4.6, N4.7	Instrument
R04	Independent check of the initiator charge: weigh D-104 before and after dosing with a high-mass interlock on XV-103, and latch the dose so it can be given only once per batch.	N3	N3.1, N3.7	Instrument / Process
R05	Organic peroxide storage: refrigerated store with independent high-temperature alarm and back-up cooling, inventory limited to one shift at D-104, supplier SADT and emergency temperatures posted, separation from the reactor bay.	N3	N3.6	Process / Safety
R06	LOPA for the runaway scenarios that rely on TSHH-101, PSHH-101, XS-101, FSLL-110 and the short-stop; set the SIL target and the proof-test interval of XV-104 and the logic solver.	N4/N5/N8	N3.1, N4.1, N5.7, N7.1, N8.1	Process safety
R07	Add a pressure transmitter with low-pressure alarm on the short-stop pot D-105 and make it an initiation permissive.	N8	N8.1, N8.2	Instrument
R08	Confirm PSV-100 / XV-108 emergency-relief sizing for the runaway case with two-phase flow (DIERS method) and the capacity of D-103 for one full relief.	N8	N8.6, N8.7	Process
R09	CW supply: low header-pressure alarm and CW supply flow on the reactor DCS; study CW failure as a common cause for all four reactors (stagger peak heat release, back-up CW pump on essential power).	N5/N6	N5.7, N5.8, N6.1	Process / Utilities
R10	Deviation alarm TE-100 vs TE-101 (1 °C) so a drifting control element is seen before the K-value or the trip margin is lost.	N4	N4.1, N4.3	Instrument
R11	Deviation alarm PT-100 vs PT-101 and a frozen-value check on PI-100, because the PD-100 end point depends on PT-100 alone.	N4/N10	N4.5, N9.4, N10.5	Instrument
R12	Double block and bleed on the nitrogen supply to D-105 (open only to pressurise the pot); consider automatic inert venting on TI-132 low with a flow limit on HIC-135.	N7/N8	N4.4, N7.4, N8.3	Process / Instrument
R13	Calculated E-101 duty check (FI-130 × ΔT against the heat balance) with a deviation alarm; include FT-130 in the calibration plan.	N7	N7.2	Instrument
R14	Car-seal open the E-101 CW block valves and add FI-130 LO alarm while E-101 is in service.	N7	N7.1, N7.2	Operations / Instrument
R15	Fouling programme: wall, baffle and E-101 cleaning interval set from the TV-110B and TI-131 trends; keep the VPC-130 anti-foam limit in service.	N4/N6/N7	N4.10, N4.11, N6.2, N6.5, N7.3, N7.5	Operations
R16	Confirm LSHH-100 uses its own level sensor (not LT-100) and add a check of LIC-100 against the load cells WI-100.	N1/N2/N4	N1.2, N2.2, N4.8, N4.9	Instrument
R17	Second shut-off valve on the injection-water line closing on LSHH-100, and an alarm "FI-103 flow with FIC-103 output at zero".	N4	N4.8	Instrument
R18	Tight shut-off block valve on LP steam to SM-101, closed while the batch reacts, with position indication on the DCS.	N5	N5.4, N5.6	Process / Instrument
R19	Valve-position feedback on TV-110B with a deviation alarm against the TIC-110 output (stuck valve detection).	N5	N5.4	Instrument
R20	Vacuum-hold test (pressure rise over 5 min) before the VCM charge; oxygen check of recovered VCM before re-use.	N1/N2/N10	N1.7, N2.4, N10.1	Operations
R21	Remotely operated isolation valve at the TK-100 outlet and VCM gas detectors at P-103, the charge floor and the agitator seal, with automatic isolation of the charge.	N2	N2.6, N4.12	Process / Instrument
R22	Make short-stop (or end point reached) a hard permissive for opening XV-105; today the page only gives a warning alarm.	N9	N9.3	Instrument
R23	Route the D-102 vent to VCM recovery, add D-102 high-level alarm and confirm D-102 holds one full batch plus margin.	N9	N9.4, N9.5, N9.7	Process
R24	Reactor opening: degas to recovery, steam-strip and evacuate until a portable VCM measurement at the manway is below the occupational limit; closed-reactor high-pressure water cleaning so that man-entry is the exception; confined-space permit with positive isolation.	N10	N9.9, N10.7, N10.8	Operations / HSE
R25	Recipe presets (FQI-101, FQI-102, initiator) downloaded from a recipe system and checked by a second person before the charge.	N1/N2/N10	N1.2, N1.3, N1.6, N2.2, N10.9	Operations
R26	Show VCM recovery status (compressor C-101 running, gasholder D-106 level) on the reactor DCS; keep vacuum-pump exhaust away from the recovery system (vent treatment only).	N7/N9/N10	N7.6, N7.7, N9.2, N9.8, N10.2	Process

No.	Recommendation	Node	Raised from	Action by
R27	Train operators on the OTS for loss of agitation, loss of cooling, initiator overcharge and hot initiation, including when to short-stop by hand.	All	N3.1, N3.5, N4.7, N5.9, N7.9, N10.4	Operations
R28	Trip-bypass management: any SIS bypass needs a permit, a time limit and MOC; bypass status shown on the DCS.	N8	N2.4, N8.9	Operations
R29	Measure the PVA charge (flow totaliser or TK-107 load cell) and make it a VCM-charge permissive.	N1	N1.6	Instrument
R30	Check valves plus the FC charge valves on every line into R-101 (water, VCM, initiator); confirm back-flow protection because R-101 runs above the water header and the storage pressure.	N1/N2/N3	N1.9, N2.7, N3.8	Process
R31	Minimum-flow recycle to TK-100 on P-103 and a double mechanical seal with barrier-pressure alarm.	N2	N2.5, N2.6	Mechanical
R32	Specification check of recovered VCM before re-use (oxygen, acetylene, HCl, water).	N2	N2.8	Laboratory
R33	Initiator lot control: assay certificate, first-in-first-out, store temperature records; reject drums that have been warm.	N3	N3.2, N3.6, N3.7	Laboratory / Operations
R34	VCM detection on the CW returns from the baffles, jacket overflow and E-101 (tube-leak detection) before the cooling tower.	N6/N7	N6.4, N7.8	Instrument
R35	Rupture disc with pressure-monitored interspace under PSV-100 to stop polymer plugging the relief path; inspect the relief line by batch count.	N8	N8.7	Mechanical
R36	Review the spurious-trip rate of the SIS initiators (e.g. 2003 voting on temperature and pressure) together with the SIL target from R06.	N8	N8.5	Instrument
R37	Degassing rate limit (ramp on HIC-105) and a knock-out pot ahead of the gasholder to catch foam and slurry.	N9	N9.1	Process
R38	Written rule for restarting the agitator after it has stopped on a batch that is still reacting (only after the short-stop has been confirmed, with full cooling), practised on the OTS.	N10	N8.4, N10.10	Operations
R39	Dedicated instrument-air receiver (with check valve) for XV-104 and XV-108 so the short-stop and emergency vent can still open after an air failure.	N8	N8.8	Instrument

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here selected deviations are run on copies of the R-101 model from the 2.0 h design state ($X \approx 26\%$) with the existing safeguards active — alarms, SIS trips, controllers in their normal modes — and **no operator action**, for 30 min. The result shows whether the safeguards act before harm and how much time the operator has. Alarms are checked every second with the page's alarm table; the live simulator is not touched.

Ref	Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out) and time	Key tags: range in 30 min	Short-stop
D1	N4	Agitation NO	Agitator motor M-101 trips (electrical)	• 00:16 SIS TRIP • 19:48 TI-132 LO	XS-101 agitator M-101 stopped while reacting at 00:15	• JI-100 0 to 0 kW • TI-101 45.5 to 57.0 °C • PI-101 6.30 to 8.66 bar g	injected
D2	N4	Agitation LESS	Impeller drive lost with the motor still running (coupling / shaft)	• 00:01 JI-100 LO	no trip	• JI-100 58 to 59 kW • TI-101 57.0 to 57.2 °C • Coarse fraction 0.0 to 16.0 %	—
D3	N5	Flow NO	Running jacket pump P-101A trips, spare P-101B not started	• 00:02 FI-110 LO • 00:36 SIS TRIP • 21:58 TI-132 LO	FSLL-110 jacket circulation below 120 m³/h while reacting at 00:36	• FI-110 0 to 365 m³/h • TI-101 46.7 to 57.1 °C • PI-101 6.53 to 8.68 bar g	injected
D4	N5	Utility NO	Cooling-water supply lost (header flow –95 %)	• 00:01 FIC-120 LO • 06:02 TIC-100 HI • 10:33 TI-131 HI • 13:23 TI-101 HI	TSHH-101 R-101 temperature high-high at 20:45	• TI-101 57.0 to 63.1 °C • PI-101 8.62 to 10.10 bar g • TV-110B 55 to 100 % open	injected
D5	N3	Flow MORE	A second fast-initiator charge reaches the reacting batch (double dose)	• 04:38 TIC-100 HI • 11:34 TI-101 HI • 17:30 SIS TRIP	TSHH-101 R-101 temperature high-high at 17:30	• TI-101 53.8 to 63.1 °C • PI-101 7.94 to 10.09 bar g • Heat release 0.03 to 9.81 MW	injected
D6	N7	Flow NO	E-101 cooling water isolated (block valve closed)	• 06:22 TIC-100 HI • 16:08 TI-101 HI • 25:30 SIS TRIP	TSHH-101 R-101 temperature high-high at 25:30	• TI-101 57.0 to 63.1 °C • PI-101 8.62 to 10.09 bar g • TV-110B 53 to 100 % open	injected
D7	N7	Composition AS WELL AS	Nitrogen passing into R-101 from the short-stop pot supply	• 24:18 TI-132 LO	no trip	• TI-132 44.3 to 54.0 °C • PI-101 8.62 to 9.62 bar g • FI-130 36 to 69 m³/h	—
D8	N4	Measurement OTHER THAN	Control element TE-100 drifts to 3 °C low over 10 min	• 10:05 TI-131 HI • 13:05 TI-101 HI	no trip	• TIC-100 (TE-100) 56.6 to 57.2 °C • TI-101 57.0 to 60.2 °C • PI-101 8.62 to 9.35 bar g	—

Ref	Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out) and time	Key tags: range in 30 min	Short-stop
D9	N5	Temperature LESS	TIC-110 output fails to 100 % (TIC-110 forced to MAN, full CW injection)	• 05:10 TIC-100 LO	no trip	• TI-101 49.8 to 57.0 °C • TV-110B 67 to 100 % open • Heat release 1.45 to 2.71 MW	—
D10	N8	Utility NO	Instrument air lost: every valve to its fail position	• 03:34 TIC-100 LO	no trip	• TI-101 48.6 to 57.0 °C • PI-101 6.83 to 8.62 bar g • FI-130 184 to 540 m³/h	—

What the runs tell the HAZOP team

- **Agitator stops (N4):** XS-101 short-stops the batch at 00:15; the trip is the first annunciation (JI-100 LO is suppressed once the motor is off). TI-101 then falls to 45.5 °C and PI-101 to 6.30 bar g. Without the impeller the short-stop mixes in minutes rather than seconds; with full cooling the batch still cools here because the stop came at 2 h ($X \approx 26\%$), well before the heat-release peak — near the gel peak the margin would be much smaller, hence R02 (essential power) and N8.4.
- **Agitation lost with the motor running (N4):** JI-100 LO at 00:01 is the only alarm and nothing trips. The controllers hold TI-101 at 57.0–57.2 °C, so temperature does not warn anyone, while the coarse fraction reaches 16.0 % in 30 min (R03).
- **Jacket pump trips (N5):** FI-110 LO at 00:02, FSLL-110 short-stops the batch at 00:36: about 34 s for the operator to start P-101B (R01).
- **Cooling water lost (N5):** FIC-120 LO at 00:01, TIC-100 HI at 06:02, TI-101 HI at 13:23, TSHH-101 at 20:45 with PI-101 at 10.10 bar g. Roughly 21 min from the loss to the trip — time for a controlled short-stop by hand, but on all four reactors at once (R09).
- **Double initiator dose (N3):** heat release climbs to 9.81 MW against 2.71 MW at design; TIC-100 HI at 04:38, TI-101 HI at 11:34, TSHH-101 at 17:30. The PSV does not lift: the trip protects, but only an independent dose check prevents it (R04).
- **E-101 cooling water isolated (N7):** TV-110B reaches 100 % at 02:48, TIC-100 HI at 06:22 and TSHH-101 trips at 25:30. At 2 h the jacket alone cannot hold the batch, so E-101 cooling is safety-relevant: N7.1 is ranked High and R14 (car-seal, FI-130 LO) follows.
- **Nitrogen passing into R-101 (N7):** the only alarm is TI-132 LO at 24:18; VPC-130 quietly raises the E-101 water from 36 to 69 m³/h and PI-101 rises to 9.62 bar g. The controller hides the loss of condenser area (R12).
- **TE-100 drifting low (N4):** TIC-100 never alarms (its own PV stays near 57 °C), while the real temperature reaches 60.2 °C; the first warnings are TI-131 HI at 10:05 and TI-101 HI at 13:05. Only the independent TE-101 shows it — hence the deviation alarm R10.
- **Jacket CW valve driven fully open (N5):** TIC-100 LO at 05:10; the batch cools to 49.8 °C and the heat release falls to 1.45 MW — the safe direction, but off grade and slower.
- **Instrument air lost (N8):** no trip; the fail-open cooling valves give full cooling (FI-130 up to 540 m³/h) and the batch cools to 48.6 °C. The fail positions are right for cooling, but XV-104 and XV-108 (fail closed) could not open if a runaway came from another cause at the same time (R39).

5.4 Follow-up and close-out

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with a justification agreed by the HAZOP leader and the plant manager). The register in section 8 is the tracking list.
- High scenarios whose protection relies on instrumented trips (runaway by loss of cooling, agitation or by initiator overcharge) go to **LOPA** to set the SIL of the short-stop function (IEC 61511) — see the Safety layers and LOPA tabs.
- The emergency relief basis (PSV-100, XV-108, D-103) is confirmed by a runaway relief study before the recipe or the initiator is changed.
- Any change — a new initiator, a faster recipe, a bigger charge, a new cleaning method — goes through **management of change**, which checks whether the HAZOP must be updated.
- Revalidate the HAZOP periodically (typically every five years) and after incidents or near misses.
- Use the simulator to train operators on the scenarios this HAZOP found and to check that the alarm responses and procedures work in time.

Appendix 1 · HAZOP worksheets

Node N1 · Demineralised water and PVA charge

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N1 · Demineralised water and PVA charge
Design intent	Charge 60 t of demineralised water at 55 °C through FIC-101 (FV-101 fail closed) and XV-101 (fail closed) until FQI-101 reaches its preset, with the PVA suspending agent from pot TK-107 (HV-107) into the evacuated R-101.	Design conditions	60 t at 55 °C, FIC-101 set point 240 m ³ /h, water header 5.0 bar g; R-101 under vacuum
Equipment	Demineralised-water header, TK-107 PVA pot, HV-107, FV-101, XV-101, charge line to R-101	Instruments	FIC-101, FQI-101 (preset), LIC-100, PI-100, WI-100
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No water reaches R-101	• XV-101 or FV-101 fails closed • demineralised-water supply lost • FQI-101 preset left at zero	Batch cannot start; cycle time lost (operability only)	• FQI-101 totaliser • batch sequence shows the charge step • VCM charge blocked until water ≥ 95 % of preset	1	3	3 L	—	—
N1.2	Flow	MORE	Too much water charged	• FQI-101 preset entered too high • totaliser miscounts • XV-101 fails to close at the preset	Reactor overfilled: after VCM and heating the vapour space vanishes, liquid reaches E-101 and the vessel can go liquid-full → hydraulic overpressure	• LSHH-100 94 %, 5 s → charge valves and injection closed • LIC-100 HI 88 % / LO 74 % • water charge refused above 85 % level • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	4	2	8 M	R16, R25	Instrument · Operations
N1.3	Flow	LESS	Less water than the recipe	• FQI-101 preset low • header pressure low and the charge stopped early	Higher solids and viscosity, lower jacket heat transfer, hotter batch, more lumps	• VCM charge permissive (water ≥ 95 % of preset) • LIC-100 LO alarm • JI-100 HI alarm	3	2	6 M	R25	Operations
N1.4	Temperature	MORE	Charge water hotter than 55 °C	Demineralised-water heater controller fault	VCM flashes when charged, pressure rises early; heat-up faster (small effect)	• PI-100 and TI-100 indication • operator	2	2	4 L	—	—
N1.5	Temperature	LESS	Charge water cold	Heater out of service	Longer heat-up on LP steam (operability)	• TI-100 indication • heat-up step in the procedure	1	3	3 L	—	—
N1.6	Composition	PART OF	Only part of the PVA reaches the reactor	• HV-107 opened late or not at all • PVA pot not refilled • wrong concentration	Unstable suspension: droplets coalesce, lumps grow, agitator power and wall fouling rise, jacket U falls; bottom valve may block at discharge	• Start-up procedure step (PVA ≥ 95 % of recipe) • JI-100 LO 150 kW / HI 420 kW	3	3	9 M	R29, R25	Instrument · Operations
N1.7	Composition	AS WELL AS	Dissolved oxygen or air with the charge	• Water not de-aerated • air drawn in through the charge line	Induction period (oxygen inhibits), inerts that later blanket E-101, polyperoxide formation	• VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated) • TI-132 LO 46 °C (inerts at the E-101 top)	2	3	6 M	R20	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.8	Composition	OTHER THAN	Wrong water quality	- Raw or softened water instead of demineralised - iron or chloride carry-over	Fouling and off-colour PVC, porosity out of specification (quality)	Laboratory checks of the demineralised water	2	2	4 L	—	—
N1.9	Flow	REVERSE	Reactor contents back into the water header	XV-101 and FV-101 passing while R-101 runs at 8.6 bar g, above the header pressure	VCM enters the demineralised-water system and is released at other users (exposure, fire)	- XV-101 and FV-101 both fail closed and closed by the sequence/SIS - header pressure alarm (utility)	4	2	8 M	R30	Process

Node N2 · VCM charge

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N2 · VCM charge: TK-100 → P-103 → FV-102 / XV-102 → R-101
Design intent	Charge 51 t of VCM at 25 °C from storage TK-100 with P-103 through FIC-102 (FV-102 fail closed) and XV-102 (fail closed) until FQI-102 reaches its preset; only with the reactor evacuated, water in, M-101 running and the jacket circulating.	Design conditions	51 t at 25 °C, FIC-102 set point 180 t/h, pump discharge 12.0 bar g; level after charge ≈ 83 %
Equipment	TK-100, P-103, FV-102, XV-102, VCM charge line	Instruments	FIC-102, FQI-102 (preset), PI-100, LIC-100, LSHH-100
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	NO	No VCM charged	- P-103 trips or is not started - XV-102 refused by a missing permissive	Batch delayed (operability)	- Charge refused with the missing permissive named - FQI-102	1	3	3 L	—	—
N2.2	Flow	MORE	Too much VCM	- FQI-102 preset too high - totaliser error	Higher level and heat per volume; liquid-full on heating → overpressure; more heat at the gel peak than the cooling can remove	- LSHH-100 94 %, 5 s → charge valves and injection closed - LIC-100 HI 88 % / LO 74 % - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	4	2	8 M	R16, R25	Instrument · Operations
N2.3	Flow	LESS	Less VCM than the recipe	- Preset low - meter error	Lower level and batch mass; smaller jacket area wetted; lower productivity	- LIC-100 LO alarm - FQI-102	2	2	4 L	—	—
N2.4	Sequence	BEFORE	VCM charged before the reactor is evacuated	- Evacuation cut short - XV-109 still open - permissive bypassed	Air and VCM in the vapour space: flammable mixture, polyperoxides, inerts that blanket E-101	- VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated) - VCM charge refused with XV-109 open	4	2	8 M	R20, R28	Operations
N2.5	Pressure	MORE	P-103 runs against a closed valve	XV-102 or FV-102 closes (preset, LSHH-100, trip) with P-103 still running	Pump overheats; seal failure → VCM leak at the pump (fire, exposure)	- Operator stops P-103 after the charge - pump seal	4	2	8 M	R31	Mechanical
N2.6	Integrity	OTHER THAN	VCM leak from the charge system	- Pump seal, flange or gasket failure - unloading hose - line hit during maintenance	Flashing VCM release: vapour cloud, flash fire or explosion; exposure to a carcinogen	- XV-102 fail closed (closed by any trip) - FV-102 fail closed - plant fire protection	5	2	10 H	R21, R31	Mechanical · Process / Instrument
N2.7	Flow	REVERSE	Reactor contents back to TK-100	P-103 stopped with XV-102/FV-102 open while R-101 is above the storage pressure	Slurry and initiator into the VCM store: polymerisation and fouling in storage, overpressure of the store	- XV-102 closes at the preset (FQI-102) - XV-102 and FV-102 fail closed	4	2	8 M	R30	Process
N2.8	Composition	AS WELL AS	Impurities in the VCM (oxygen, acetylene, HCl, water)	Recovered VCM returned to storage out of specification	Induction period, slower batch, off-grade K-value, polyperoxide hazard with oxygen	Laboratory analysis of fresh VCM	2	3	6 M	R32	Laboratory

Node N3 · Initiator charge

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N3 · Initiator charge: D-104 → P-104 → XV-103
Design intent	With the VCM charged and TI-100 ≥ 54 °C, dose the peroxide initiator mix (fast A, half-life 1.5 h, and slow B, 12 h, at 57 °C) from the chilled pot D-104 with P-104 through XV-103 in 60 s; the reaction time starts.	Design conditions	One dose per batch in 60 s; D-104 kept chilled; initiation permissive TI-100 ≥ 54 °C and FQI-102 ≥ 95 %
Equipment	D-104 (chilled), P-104 dosing pump, XV-103, initiator line; peroxide store	Instruments	Batch sequence (initiator permissive), TI-100, XI-100
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	MORE	More initiator than the recipe	- Weighing error - pot flushed twice (second dose) - wrong concentration	Heat release beyond the cooling capacity: temperature and pressure rise, runaway, relief of VCM	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	5	2	10 H	R04, R06, R27	Instrument / Process · Operations · Process safety
N3.2	Flow	LESS	Less initiator activity	- Aged or warm-stored lot - under-weighed dose	Slow batch, long cycle, more VCM left at the end (operability, recovery load)	- XI-100 against the reference batch - heat-balance display	2	3	6 M	R33	Laboratory / Operations
N3.3	Flow	NO	No initiator	- P-104 fails - XV-103 does not open - empty pot	No reaction; the batch sits hot (operability)	- XI-100 stays at zero - operator	1	3	3 L	—	—
N3.4	Timing	EARLY	Initiator before the batch is ready	Dosed at low temperature or before the VCM charge ends	Reaction starts during the charge, level and heat load not as designed	Initiation permissive: FQI-102 ≥ 95 % and TI-100 ≥ 54 °C	3	1	3 L	—	—
N3.5	Timing	LATE	Initiator dosed while the jacket is still hot from heat-up	- Steam heat-up not stopped in time - TIC-110 left on high jacket temperature	Temperature overshoot in the first hour as the fast initiator starts: off-grade K-value, approach to the trip	- TIC-110 HI 75 °C alarm while the initiator is in (first hour) - TIC-100 HI 58 °C / LO 56 °C - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	3	2	6 M	R27	Operations
N3.6	Temperature	MORE	Initiator pot or store warms up	- Refrigeration failure - drum left at ambient - heat from the reactor bay	Self-accelerating decomposition of the peroxide: pot overpressure, fire, toxic fumes	- Chilled pot and store - small pot inventory - temperature indication at the store	4	2	8 M	R05, R33	Laboratory / Operations · Process / Safety

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.7	Composition	OTHER THAN	Wrong initiator	Wrong drum or grade picked (faster peroxide)	Very fast start, heat peak early and above the cooling capacity	• TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open • PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	5	1	5 M	R04, R33	Instrument / Process · Laboratory / Operations
N3.8	Flow	REVERSE	Reactor contents into D-104	XV-103 left open after dosing with P-104 stopped (R-101 at 8.6 bar g)	VCM and slurry contaminate the peroxide pot → decomposition, pot overpressure	• XV-103 fail closed and closed after the 60-s dose • dosing pump positive displacement	4	1	4 L	R30	Process

Node N4 · R-101 during the reaction

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N4 · R-101 during the reaction (temperature, pressure, level, agitation, composition)
Design intent	Hold 57 °C (grade K 67, band ± 1 °C) at about 8.6 bar g with the level at 83 % (LIC-100 → FIC-103 shrinkage water) and M-101 running, for ≈ 5.3 h until PD-100 sees a 1.8 bar drop below the running pressure; heat release peaks at ≈ 3.4 MW near 72 % conversion.	Design conditions	57.0 °C, 8.62 bar g, level 82.9 %, M-101 290 kW, heat release 2.71 MW at 2.0 h
Equipment	R-101 (140 m ³ , design 15.2 bar g / 93 °C), agitator M-101 (330 kW), FV-103 injection	Instruments	TIC-100 (TE-100), TI-101 / TSHH-101 (TE-101), PI-100 (PT-100), PI-101 / PSHH-101 (PT-101), LIC-100, LSHH-100, WI-100, JI-100, XS-101, XI-100, PD-100, KX-100
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Temperature	MORE	Reactor hotter than 57 °C	- Heat release above the removal capacity (gel-effect peak, warm CW, fouled wall, extra initiator) - loss of any cooling handle	K-value off grade, pressure rises with the VCM vapour pressure, runaway to relief	- TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	5	2	10 H	R06, R10	Instrument · Process safety
N4.2	Temperature	LESS	Reactor colder than 57 °C	- Over-cooling (CW valve fails open, TIC-110 output high) - steam unavailable after a stop	Slow batch, K-value high (off grade); safe direction for the reaction	- TIC-100 HI 58 °C / LO 56 °C - TIC-100 → TIC-110 cascade, split range TV-110A steam (fail closed) / TV-110B CW (fail open)	2	3	6 M	—	—
N4.3	Measurement	OTHER THAN	Control element TE-100 reads low	Thermocouple drift or poor thermowell contact	TIC-100 holds the wrong temperature: the batch runs hot, K-value low, trip margin shrinks without a TIC-100 alarm	- TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	3	3	9 M	R10	Instrument
N4.4	Pressure	MORE	Pressure above the VCM vapour pressure	- Inerts (nitrogen passing, poor evacuation) collecting in E-101 - temperature high	E-101 duty falls, pressure and temperature rise; relief if uncontrolled	- PI-100 HI 11 bar g - TI-132 LO 46 °C (inerts at the E-101 top) - HIC-135 inert vent - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop	4	2	8 M	R12	Process / Instrument
N4.5	Measurement	OTHER THAN	PT-100 frozen or wrong	Transmitter fault, plugged impulse line	PD-100 end point not seen: batch runs on in the pressure-drop phase (quality, cycle time) or is stopped early (VCM load on recovery)	- PI-101 (independent SIS transmitter) on the display - XI-100 conversion estimate	2	3	6 M	R11	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.6	Agitation	NO	Agitator stops	- M-101 motor trip - power failure	Suspension separates within a minute: VCM droplets coalesce, jacket heat transfer collapses, hot spots and lumps; runaway without short-stop	- XS-101 agitator stopped while reacting, 15 s → short-stop - JI-100 LO 150 kW / HI 420 kW	5	2	10 H	R02, R03	Electrical · Instrument
N4.7	Agitation	LESS	Agitation lost with the motor running	- Coupling or shaft failure - impeller loose	As above but XS-101 does not see it (motor shows RUN); lumps grow; heat transfer falls	- JI-100 LO 150 kW / HI 420 kW - TIC-100 via more jacket CW and E-101 water	5	2	10 H	R03, R27	Instrument · Operations
N4.8	Level	MORE	Level above 83 %	- FV-103 passing - LT-100 radar reads low - LIC-100 set point high	Vapour space shrinks, foam and liquid into E-101 and the vapour line; liquid-full → hydraulic overpressure	- LIC-100 HI 88 % / LO 74 % - FIC-103 HI 8 m³/h - LSHH-100 94 %, 5 s → charge valves and injection closed - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	4	2	8 M	R16, R17	Instrument
N4.9	Level	LESS	Level below 83 %	- Injection water stopped or isolated - LT-100 reads high	Wall above the liquid dries: polymer build-up, less jacket area (the injection exists to keep the wall wetted)	- LIC-100 HI 88 % / LO 74 % - WI-100 load cells	2	3	6 M	R16	Instrument
N4.10	Composition	AS WELL AS	Foam and polymer carried into E-101	Boil-up (E-101 water) raised too early in the batch	E-101 tubes and vapour line foul, condenser duty lost for later batches	- VPC-130 anti-foam limit (12 m³/h per % XI-100) - XA-133 foam alarm	3	3	9 M	R15	Operations
N4.11	Heat transfer	LESS	Polymer build-up on the reactor wall	- Wall scale from previous batches - poor cleaning	Jacket U falls; TV-110B opens further and the cooling margin at the peak shrinks	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C - wall cleaning between batches	3	3	9 M	R15	Operations
N4.12	Integrity	OTHER THAN	VCM leak from R-101	- Agitator seal failure - manway, bottom-valve or flange gasket failure	Flashing VCM release at 57 °C / 8.6 bar g: vapour cloud, flash fire or explosion; exposure to a carcinogen	- Agitator seal with barrier fluid (assumed) - XV-106 fail closed - ESD HS-001 closes the charges - plant fire protection	5	2	10 H	R21	Process / Instrument

Node N5 · Jacket tempered-water loop with CW and LP-steam injection

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N5 · Jacket tempered-water loop with CW and LP-steam injection
Design intent	Circulate tempered water (P-101A running, P-101B spare) through the jacket; TIC-110 (slave of TIC-100) sets the jacket inlet by split range: 0–50 % TV-110A LP steam (fail closed) into mixer SM-101, 50–100 % TV-110B CW injection (fail open); the surplus overflows to CW return.	Design conditions	FI-110 473 m³/h, jacket in 38.2 °C / out 40.9 °C, TV-110B 53 % open, jacket duty 1.50 MW
Equipment	P-101A/B, SM-101, TV-110A, TV-110B, jacket (5.2 bar g), loop piping	Instruments	TIC-110 (TE-110), TI-111, FI-110 / FSLL-110, TI-120
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No jacket circulation	- P-101A trips and P-101B is not started - power failure - suction valve closed	Jacket cooling lost: temperature and pressure rise, runaway	- FI-110 LO 300 m³/h - FSLL-110 jacket circulation below 120 m³/h while reacting, 30 s → short-stop - spare P-101B (manual start)	5	2	10 H	R01, R02	Electrical · Instrument / Operations
N5.2	Flow	LESS	Low jacket circulation	- Pump wear - strainer fouled - air in the loop	Lower jacket heat transfer, less cooling margin at the peak	- FI-110 LO 300 m³/h - TIC-110 compensates with more CW	3	2	6 M	R01	Instrument / Operations
N5.3	Flow	REVERSE	Flow back through the idle pump	P-101B discharge check valve passing	Recirculation through the spare, less jacket flow, spare turns backwards	FI-110 LO 300 m³/h	2	2	4 L	—	—
N5.4	Temperature	MORE	Jacket inlet hotter than demanded	- TV-110B sticks - TIC-110 in MAN - LP steam passing TV-110A	Less heat removed: temperature rises until TIC-100 finds another handle or the trip acts	- TIC-100 → TIC-110 cascade, split range TV-110A steam (fail closed) / TV-110B CW (fail open) - TIC-100 HI 58 °C / LO 56 °C - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	4	2	8 M	R18, R19	Instrument · Process / Instrument
N5.5	Temperature	LESS	Jacket inlet colder than demanded	- TIC-110 output fails high - TV-110B fails open (fail position)	Batch over-cooled: reaction slows, K-value high (safe direction)	TIC-100 HI 58 °C / LO 56 °C	2	3	6 M	—	—
N5.6	Flow	AS WELL AS	Steam enters the loop while the batch reacts	- TV-110A passing - split-range calibration error	Hidden heat load: more CW needed for the same temperature, margin at the peak lost	- TIC-110 corrects with TV-110B - TV-110B position on the DCS	3	3	9 M	R18	Process / Instrument
N5.7	Utility	NO	Cooling-water supply lost	CW pump trip or header failure (common to all four reactors)	Jacket, baffles and E-101 all lose cooling: runaway on every reacting batch	- FIC-120 LO 15 m³/h - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	5	2	10 H	R09, R06	Process / Utilities · Process safety
N5.8	Utility	OTHER THAN	Cooling water warmer than 32 °C	- Cooling-tower performance - hot afternoon	Less capacity at the gel peak; TV-110B and E-101 at their limits	- TI-120 HI 35 °C (CW supply) - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 %	3	3	9 M	R09	Process / Utilities

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.9	Control	OTHER THAN	Temperature cascade broken	- TIC-110 left in AUTO/MAN after heat-up - TIC-100 in MAN	Reactor temperature no longer controlled; drift with the heat release	- TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101)	3	3	9 M	R27	Operations
N5.10	Measurement	OTHER THAN	TE-110 jacket-inlet element reads high	Element drift	Jacket runs a little colder than displayed; TIC-100 trims it (small effect)	- TIC-100 master corrects - TI-111 comparison	2	2	4 L	—	—
N5.11	Pressure	MORE	Jacket overpressure	Steam valve fully open with the overflow blocked	Jacket (5.2 bar g) overpressure, leak of hot water (burns)	- Open overflow to CW return - LP steam pressure below the jacket design	3	1	3 L	—	—

Node N6 · Baffle cooling water FIC-120

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N6 · Baffle cooling water FIC-120
Design intent	Cool the four internal baffles with CW through FIC-120 (FV-120 fail open) at 80 m³/h; return monitored by TI-121.	Design conditions	FIC-120 80 m³/h, return 36.3 °C, baffle duty 0.40 MW
Equipment	FV-120, four baffles, CW supply and return	Instruments	FIC-120, TI-121, TI-120
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	NO	No baffle water	- CW supply lost - baffle inlet blocked - controller fault closing FV-120	About 0.4 MW of cooling lost; TIC-100 shifts the load to the jacket and E-101	- FIC-120 LO 15 m³/h - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C	3	2	6 M	R09	Process / Utilities
N6.2	Flow	LESS	Low baffle water	Strainer or baffle tubes scaled	Less baffle duty (small), hotter return	- FIC-120 opens FV-120 - TI-121	2	3	6 M	R15	Operations
N6.3	Flow	MORE	High baffle water	FV-120 fails open (fail position)	Slightly more cooling (safe direction)	TIC-100 corrects with the jacket	1	3	3 L	—	—
N6.4	Integrity	OTHER THAN	Baffle tube leak	Corrosion, fatigue	R-101 (8.6 bar g) is above the CW pressure: VCM enters the CW return and is released at the cooling tower (exposure, fire)	None specific	4	2	8 M	R34	Instrument
N6.5	Heat transfer	LESS	Polymer on the baffles	Build-up from previous batches	Baffle duty falls over the campaign	- Cleaning between batches - TI-121 trend	2	3	6 M	R15	Operations
N6.6	Temperature	MORE	Baffle return hot	Low flow with high heat load	Scaling on the CW side accelerates	TI-121 indication	2	2	4 L	—	—

Node N7 · Reflux condenser E-101 and inert vent

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N7 · Reflux condenser E-101 and inert vent
Design intent	Condense VCM boiling off the slurry in E-101 on CW (FIC-130, FV-130 fail open) with the condensate running back; VPC-130 moves the E-101 water to hold the jacket CW valve TV-110B near 50 % (anti-foam limit 12 m³/h per % XI-100); inerts vented by HIC-135 (fail closed) to VCM recovery at 1.25 bar a.	Design conditions	E-101 duty 0.97 MW, FIC-130 36 m³/h, TI-131 55.0 °C, TI-132 54.0 °C
Equipment	E-101 (vertical, VCM in tubes), FV-130, HV-135, vapour line, vent to recovery	Instruments	FIC-130 (FT-130), VPC-130, TI-131, TI-132, HIC-135
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Flow	NO	No cooling water to E-101	- Block valve on the E-101 CW closed by mistake - CW supply lost	E-101 duty lost; the jacket alone cannot hold the peak: temperature and pressure rise to the trip	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % (no effect when the water is blocked) - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	5	2	10 H	R14, R06	Operations / Instrument · Process safety
N7.2	Flow	LESS	Less E-101 water than displayed	- FT-130 reads high - partly closed block valve	E-101 does less than the display says; TV-110B carries more; margin shrinks	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TI-131 HI 58 °C (E-101 CW return)	3	3	9 M	R13, R14	Instrument · Operations / Instrument
N7.3	Flow	MORE	Too much boil-up early in the batch	- FIC-130 in AUTO at a high set point - anti-foam limit bypassed	Foaming: PVC carried into the tubes and vapour line (fouling, blockage)	- VPC-130 anti-foam limit - XA-133 foam alarm	3	2	6 M	R15	Operations
N7.4	Composition	AS WELL AS	Inerts in E-101	- Nitrogen passing from the short-stop pot supply - poor evacuation - air in the charge	Inerts blanket the tubes: duty falls, pressure rises, TV-110B opens further	- TI-132 LO 46 °C (inerts at the E-101 top) - PI-100 HI 11 bar g - HIC-135 inert vent	4	2	8 M	R12	Process / Instrument
N7.5	Heat transfer	LESS	E-101 fouled	- PVC deposits in the tubes (foam carry-over) - CW-side scale	Condenser duty falls; FIC-130 needs more water; less margin at the peak	- TI-131 HI 58 °C (E-101 CW return) - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 %	3	3	9 M	R15	Operations
N7.6	Flow	MORE	Inert vent left open	HIC-135 left at a high output after venting	VCM vented continuously to recovery: recovery load, less monomer in the batch	- FI-135 indication - operator	2	3	6 M	R26	Process
N7.7	Pressure	MORE	Recovery back-pressure high	- Recovery compressor C-101 tripped - gasholder full	Inerts cannot be vented; degassing slows (see N9)	- PI-100 - recovery operator	3	2	6 M	R26	Process
N7.8	Integrity	OTHER THAN	E-101 tube leak	Corrosion or vibration	VCM (reactor side at higher pressure) into the CW return and the cooling tower	None specific	4	2	8 M	R34	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.9	Control	OTHER THAN	E-101 not on valve-position control	VPC-130 in MAN or FIC-130 not in cascade	E-101 fixed: the jacket takes all swings and TV-110B saturates at the peak	• TIC-100 HI 58 °C / LO 56 °C • TV-110B on the DCS	3	3	9 M	R27	Operations

Node N8 · Short-stop, emergency vent and relief

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N8 · Short-stop, emergency vent and relief
Design intent	On demand (operator, or SIS on TSHH-101 / PSHH-101 / XS-101 / FSLL-110) open XV-104 to inject the short-stop from the N ₂ -pressurised pot D-105 into R-101, where the agitator mixes it within a minute; ESD also opens the emergency vent XV-108; PSV-100 at 15 bar g protects the vessel.	Design conditions	Trips: 63 °C, 12.5 bar g, agitator stopped 15 s, circulation < 120 m ³ /h 30 s; PSV 15 bar g; vessel 15.2 bar g / 93 °C
Equipment	D-105, XV-104, XV-108, PSV-100, dump / emergency vent tank D-103, SIS logic solver	Instruments	TSHH-101, PSHH-101, XS-101, FSLL-110, LSHH-100, HS-001 ESD
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Flow	NO	Short-stop does not enter on demand	- XV-104 fails to open - D-105 empty - line plugged	Runaway continues to relief	- Operator ESD HS-001 - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C) - XV-104 proof test	5	1	5 M	R06, R07	Instrument · Process safety
N8.2	Flow	PART OF	Only part of the short-stop enters	D-105 pressure low (nitrogen supply valve closed, pot leaking)	Reaction slowed but not stopped; temperature may recover and rise again	- Operator can inject again from a refilled pot - PSV-100	5	1	5 M	R07	Instrument
N8.3	Flow	AS WELL AS	Nitrogen into R-101 from D-105	Pressurising valve passing while the pot is connected	Inerts blanket E-101 (see N7.4)	- TI-132 LO 46 °C (inerts at the E-101 top) - PI-100 HI 11 bar g	4	2	8 M	R12	Process / Instrument
N8.4	Timing	LATE	Short-stop injected when the agitator has already stopped	- Power failure stops M-101 - short-stop by hand after the stop	Short-stop does not mix (minutes instead of seconds): local reaction goes on in the separated mass	- XS-101 agitator stopped while reacting, 15 s → short-stop (injects at once) - emergency vent XV-108	5	2	10 H	R02, R38	Electrical · Operations
N8.5	Flow	MORE	Short-stop injected without need	- Spurious trip - operator error	Batch killed at low conversion: off-grade, VCM to recover (operability)	- Independent trip transmitters - trip first-out display	2	3	6 M	R36	Instrument
N8.6	Pressure	MORE	Relief case reached	- Runaway not stopped by the short-stop - external fire	PSV-100 lifts: two-phase VCM/slurry to D-103; release if D-103 overfills	- PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C) - D-103 catch tank	5	1	5 M	R08	Process
N8.7	Flow	NO	Relief path blocked	Polymer plugs the PSV-100 inlet or outlet	Vessel overpressure beyond design: rupture	- PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - inspection of the relief line	5	1	5 M	R35, R08	Mechanical · Process
N8.8	Utility	NO	Instrument air lost	Air compressor trip, header break	XV-104 and XV-108 fail closed and cannot open on demand; cooling valves fail open (safe side)	- TV-110B, FV-120, FV-130 fail open (full cooling) - charge valves fail closed	3	2	6 M	R39	Instrument
N8.9	Safety system	OTHER THAN	SIS initiator bypassed	Bypass left in after testing or during a transmitter fault	The runaway protection is missing when needed	Administrative control only	5	2	10 H	R28	Operations

Node N9 · Degassing to VCM recovery and discharge to D-102

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N9 · Degassing to VCM recovery and discharge to D-102
Design intent	After the short-stop, degas unreacted VCM through XV-105 and HIC-105 / PV-105 (fail closed) to recovery (gasholder D-106 at 1.25 bar a, compressor C-101) to below 2 bar g, pad with N ₂ through HV-141 / PCV-141 (1.5 bar g), then discharge the slurry through XV-106 (fail closed) to blowdown tank D-102 with the agitator running.	Design conditions	Recovery at 1.25 bar a; discharge permitted below 2.0 bar g; N ₂ pad 1.5 bar g; batch ≈ 118 t
Equipment	XV-105, PV-105, recovery header, D-106, C-101, HV-141/PCV-141, XV-106, D-102, P-102	Instruments	HIC-105, FI-105, PI-100, PD-100, WI-100, FI-140
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Flow	MORE	Degassing too fast	HIC-105 opened fully at once	Foaming: slurry and PVC carried into the recovery line, gasholder and compressor fouled	• Operator throttles HIC-105 • FI-105 indication	3	3	9 M	R37	Process
N9.2	Flow	LESS	Degassing stalls	• Recovery compressor tripped • gasholder full (back-pressure)	Pressure stays high; cycle time lost; temptation to discharge early	• Discharge interlock (PI-100 below 2.0 bar g) • PI-100	3	2	6 M	R26	Process
N9.3	Timing	EARLY	Degassing a batch that is still reacting	XV-105 opened before the short-stop	Monomer vented from a live batch: evaporative cooling then foaming; recovery overloaded	Warning alarm "XV-105 OPEN" while reacting	3	2	6 M	R22	Instrument
N9.4	Sequence	BEFORE	Discharge before degassing is complete	• Interlock bypassed • PI-100 reads low	VCM flashes in D-102 and escapes at its vent: exposure and flammable atmosphere	XV-106 discharge blocked above 2.0 bar g and with the agitator stopped	4	2	8 M	R23, R11	Instrument · Process
N9.5	Flow	OTHER THAN	Bottom valve opened on a reacting batch	• Wrong reactor selected in the field • XV-106 opened by mistake	Reacting slurry at 8.6 bar g into D-102: massive VCM flash and release	• XV-106 discharge blocked above 2.0 bar g and with the agitator stopped • XV-106 fail closed	5	1	5 M	R23	Process
N9.6	Agitation	NO	Agitator stopped during discharge	M-101 tripped or stopped early	Slurry settles; bottom valve and line block; manual clearing later (exposure)	XV-106 discharge blocked above 2.0 bar g and with the agitator stopped	2	2	4 L	—	—
N9.7	Level	MORE	D-102 overfilled	• Previous batch not transferred • P-102 stopped	Slurry overflow at D-102, VCM release from the slurry	D-102 level indication (local)	3	2	6 M	R23	Process
N9.8	Flow	REVERSE	Recovery gas back into R-101	XV-105 left open with R-101 below the recovery pressure	Gas from other reactors enters R-101 (inerts, VCM in an opened reactor)	XV-105 closed by procedure below 2 bar g	2	2	4 L	R26	Process
N9.9	Sequence	AFTER	N ₂ pad left open after R-101 is empty	HV-141 not closed at the end	Nitrogen fills the next batch (inerts in E-101); N ₂ in an opened reactor (asphyxiation)	End-of-batch procedure step (HV-141 closed)	3	2	6 M	R24	Operations / HSE

Node N10 · Procedures

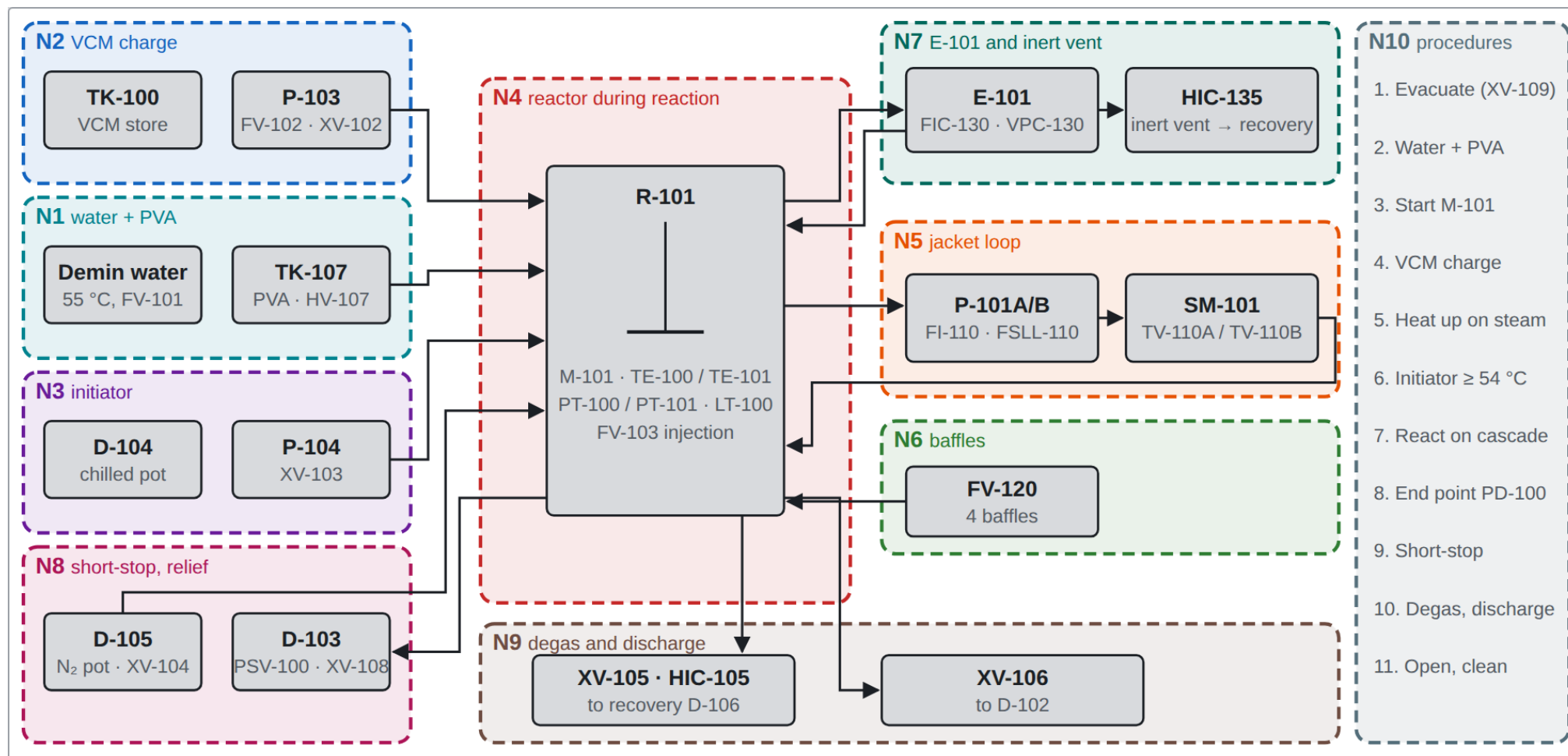
Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Node	N10 · Procedures: evacuation, charging, initiation, end point, opening the reactor
Design intent	Batch sequence: evacuate with VP-101 (XV-109) → water + PVA → agitator → VCM → heat up on steam → initiator at ≥ 54 °C → reaction on cascade → end point → short-stop → degas → discharge; between batches clean and, when needed, open the reactor.	Design conditions	Procedures in the Exercises tab (cold start, end of batch); reactor opening by permit
Equipment	Batch sequence, VP-101/XV-109, operator actions, reactor manway	Instruments	Batch & SIS display, PI-100, PD-100, FQI-101/102, TIC-100, TIC-110
Drawing	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Document	PVC-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N10.1	Evacuation	LESS	Reactor not fully evacuated	- Vacuum pump weak - XV-109 closed too early	Air left: induction period, inerts blanket E-101, oxygen in the VCM system	VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated)	3	2	6 M	R20	Operations
N10.2	Evacuation	OTHER THAN	Vacuum pump run on a reactor containing VCM	XV-109 opened with VCM still in R-101 (after degassing, cleaning)	VCM through VP-101 to the vent: exposure and flammable mixture at the vent	XV-109 refused above 1.0 bar g	4	2	8 M	R26	Process
N10.3	Charging	BEFORE	VCM before the water	Sequence step skipped	VCM without the suspension medium: no heat sink, poor mixing at initiation	VCM permissive: water ≥ 95 % of preset	3	1	3 L	—	—
N10.4	Initiation	EARLY	Initiation while the batch is still being heated hard	Steam heat-up not stopped before dosing	Overshoot above 58 °C in the first hour, off-grade K-value, approach to the trip	- TIC-110 HI 75 °C alarm while the initiator is in (first hour) - TIC-100 HI 58 °C / LO 56 °C	4	2	8 M	R27	Operations
N10.5	End point	LATE	End point missed	- PD-100 not watched - PT-100 frozen	Batch runs on in the pressure-drop phase: quality drift, long cycle	- PD-100 END alarm - PI-101 comparison	2	3	6 M	R11	Instrument
N10.6	End point	EARLY	Short-stop at low conversion	Operator short-stops on a false indication	Much VCM to degas; recovery overloaded; low yield	- PD-100 display - XI-100	2	2	4 L	—	—
N10.7	Opening	AFTER	Reactor opened before VCM is removed	- Degassing or steam-out incomplete - VCM trapped in polymer build-up	Operators exposed to VCM, a carcinogen with a very low occupational limit; flammable atmosphere at the manway	- Permit to work with gas test - respiratory protection	4	3	12 H	R24	Operations / HSE
N10.8	Opening	OTHER THAN	Man-entry for cleaning	- Scraping build-up by hand - inspection	Confined space: VCM, nitrogen (asphyxiation), agitator movement, fall	- Confined-space permit - positive isolation and lock-out of M-101 - standby person	4	2	8 M	R24	Operations / HSE
N10.9	Recipe	OTHER THAN	Wrong presets for the batch	Grade change, typing error in FQI presets or initiator amount	Over- or under-charge (see N1.2, N2.2, N3.1)	Operator check against the recipe sheet	3	2	6 M	R25	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N10.10	Restart	OTHER THAN	Agitator restarted on a stopped, still-reacting batch	Motor reset and restarted without confirming the short-stop	Sudden mixing of hot, unreacted monomer: heat surge beyond the cooling, pressure rise	• TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open • PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	4	2	8 M	R38	Operations

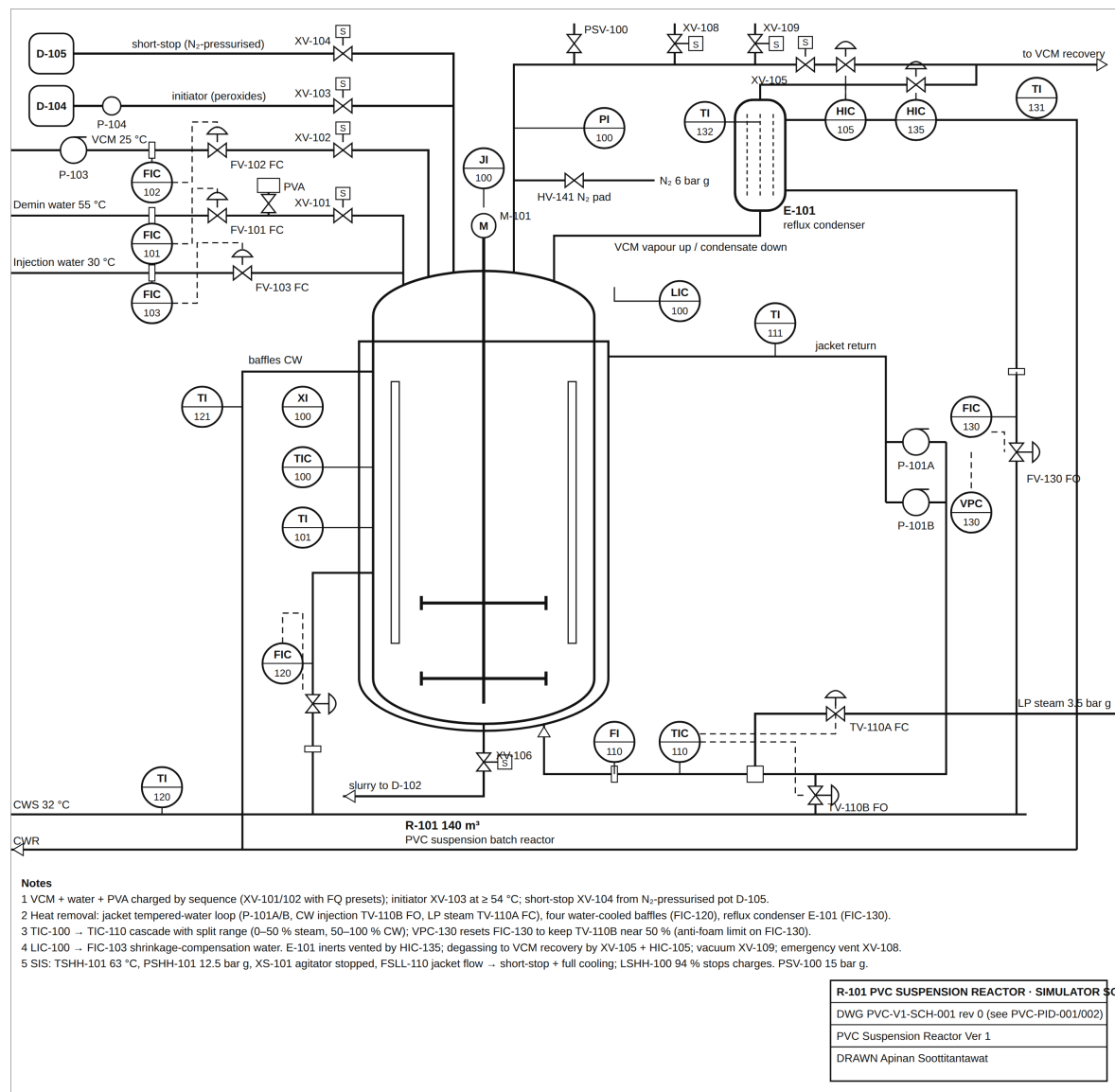
Appendix 2 · HAZOP documents (drawings with node boundaries)

A2-1 · HAZOP node map



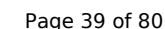
A2-1. HAZOP node map of R-101 (nodes N1-N9; N10 = procedures).

A2-2 · Simulator P&ID (reference drawing of the DCS console and the exercises)

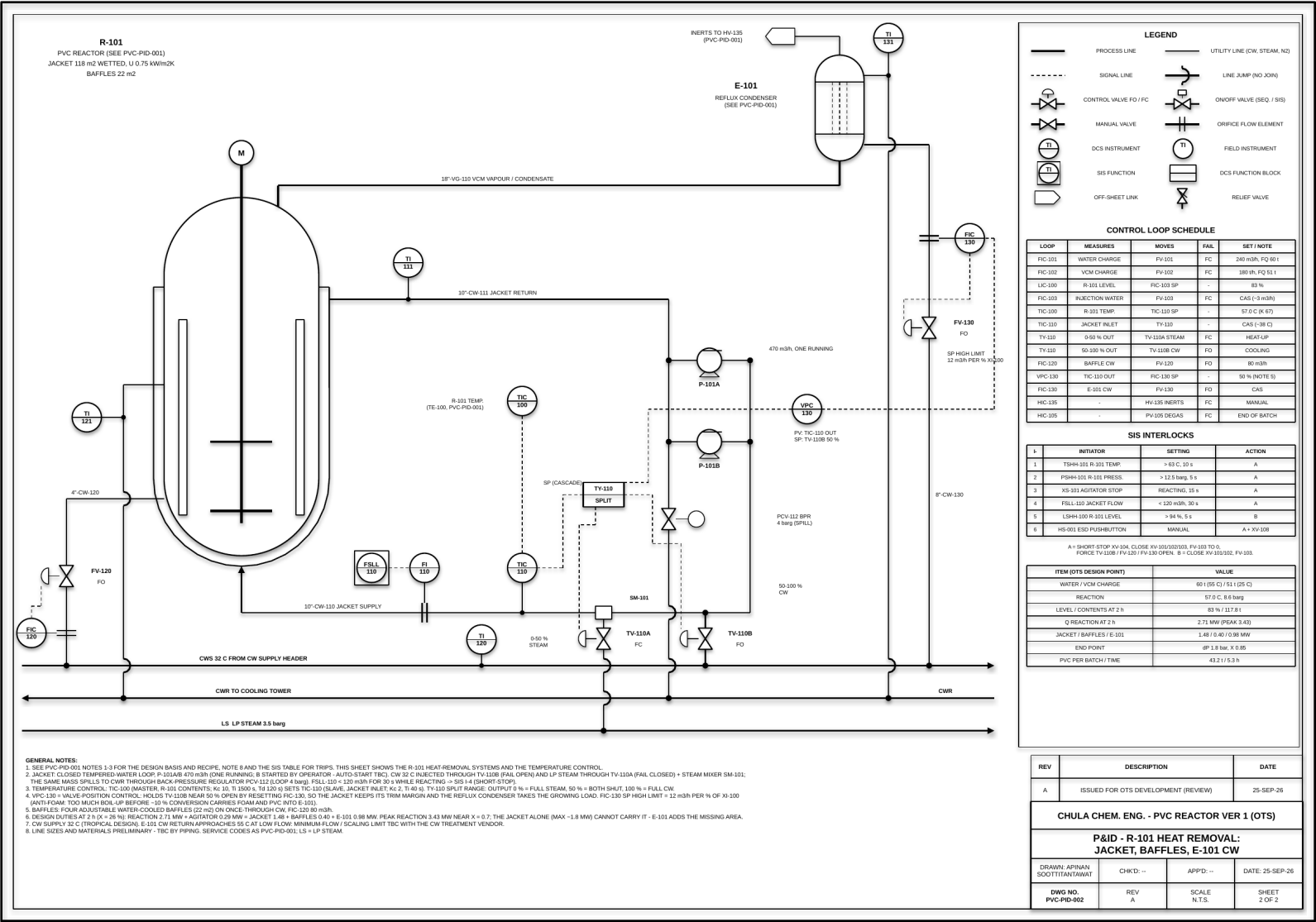


The P&ID tab drawing of R-101 as used by the DCS console, the decks and the exercises.

P&ID sheet 1 of 2 (EPC standard, alternative version on the P&ID tab).



A2-4 · PVC-PID-002 (heat removal: jacket loop, baffles, E-101 cooling water)



P&ID sheet 2 of 2 (EPC standard, alternative version on the P&ID tab).

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (39 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N5 · Jacket tempered-water loop with CW and LP-steam injection	Action party	Instrument / Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.1	Flow NO : No jacket circulation	• P-101A trips and P-101B is not started • power failure • suction valve closed	Jacket cooling lost: temperature and pressure rise, runaway	• FI-110 LO 300 m³/h • FSLL-110 jacket circulation below 120 m³/h while reacting, 30 s → short-stop • spare P-101B (manual start)	S5 L2 10 High
N5.2	Flow LESS : Low jacket circulation	• Pump wear • strainer fouled • air in the loop	Lower jacket heat transfer, less cooling margin at the peak	• FI-110 LO 300 m³/h • TIC-110 compensates with more CW	S3 L2 6 Medium

HAZOP recommendation

Auto-start the spare jacket pump P-101B on P-101A trip or FI-110 low, completing well inside the FSLL-110 delay; test the change-over monthly.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N4/N8 · R-101 during the reaction / Short-stop, emergency vent and relief	Action party	Electrical
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.6	Agitation NO : Agitator stops	• M-101 motor trip • power failure	Suspension separates within a minute: VCM droplets coalesce, jacket heat transfer collapses, hot spots and lumps; runaway without short-stop	• XS-101 agitator stopped while reacting, 15 s → short-stop • JI-100 LO 150 kW / HI 420 kW	S5 L2 10 High
N5.1	Flow NO : No jacket circulation	• P-101A trips and P-101B is not started • power failure • suction valve closed	Jacket cooling lost: temperature and pressure rise, runaway	• FI-110 LO 300 m³/h • FSLL-110 jacket circulation below 120 m³/h while reacting, 30 s → short-stop • spare P-101B (manual start)	S5 L2 10 High
N8.4	Timing LATE : Short-stop injected when the agitator has already stopped	• Power failure stops M-101 • short-stop by hand after the stop	Short-stop does not mix (minutes instead of seconds): local reaction goes on in the separated mass	• XS-101 agitator stopped while reacting, 15 s → short-stop (injects at once) • emergency vent XV-108	S5 L2 10 High

HAZOP recommendation

Put agitator M-101 and P-101A/B on the essential (emergency) power supply and prove by test that the short-stop is mixed after a power failure (P&ID note "agitator on emergency power" is still TBC).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N4 · R-101 during the reaction	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.6	Agitation NO : Agitator stops	• M-101 motor trip • power failure	Suspension separates within a minute: VCM droplets coalesce, jacket heat transfer collapses, hot spots and lumps; runaway without short-stop	• XS-101 agitator stopped while reacting, 15 s → short-stop • JI-100 LO 150 kW / HI 420 kW	S5 L2 10 High
N4.7	Agitation LESS : Agitation lost with the motor running	• Coupling or shaft failure • impeller loose	As above but XS-101 does not see it (motor shows RUN); lumps grow; heat transfer falls	• JI-100 LO 150 kW / HI 420 kW • TIC-100 via more jacket CW and E-101 water	S5 L2 10 High

HAZOP recommendation

Detect loss of agitation independently of the motor: impeller shaft-speed switch, with "low power while running" (JI-100) taken to LOPA as a candidate SIS initiator.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N3 · Initiator charge	Action party	Instrument / Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow MORE : More initiator than the recipe	• Weighing error • pot flushed twice (second dose) • wrong concentration	Heat release beyond the cooling capacity: temperature and pressure rise, runaway, relief of VCM	• VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % • TIC-100 HI 58 °C / LO 56 °C • TI-101 HI 60 °C (independent TE-101) • TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open • PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L2 10 High
N3.7	Composition OTHER THAN : Wrong initiator	Wrong drum or grade picked (faster peroxide)	Very fast start, heat peak early and above the cooling capacity	• TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open • PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L1 5 Medium

HAZOP recommendation

Independent check of the initiator charge: weigh D-104 before and after dosing with a high-mass interlock on XV-103, and latch the dose so it can be given only once per batch.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N3 · Initiator charge	Action party	Process / Safety
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.6	Temperature MORE : Initiator pot or store warms up	- Refrigeration failure - drum left at ambient - heat from the reactor bay	Self-accelerating decomposition of the peroxide: pot overpressure, fire, toxic fumes	- Chilled pot and store - small pot inventory - temperature indication at the store	S4 L2 8 Medium

HAZOP recommendation

Organic peroxide storage: refrigerated store with independent high-temperature alarm and back-up cooling, inventory limited to one shift at D-104, supplier SADT and emergency temperatures posted, separation from the reactor bay.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N4/N5/N8 · R-101 during the reaction / Jacket tempered-water loop with CW and LP-steam injection / Short-stop, emergency vent and relief	Action party	Process safety
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow MORE : More initiator than the recipe	- Weighing error - pot flushed twice (second dose) - wrong concentration	Heat release beyond the cooling capacity: temperature and pressure rise, runaway, relief of VCM	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop - XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L2 10 High
N4.1	Temperature MORE : Reactor hotter than 57 °C	- Heat release above the removal capacity (gel-effect peak, warm CW, fouled wall, extra initiator) - loss of any cooling handle	K-value off grade, pressure rises with the VCM vapour pressure, runaway to relief	- TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop - XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L2 10 High
N5.7	Utility NO : Cooling-water supply lost	CW pump trip or header failure (common to all four reactors)	Jacket, baffles and E-101 all lose cooling: runaway on every reacting batch	- FIC-120 LO 15 m³/h - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop - XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S5 L2 10 High
N7.1	Flow NO : No cooling water to E-101	- Block valve on the E-101 CW closed by mistake - CW supply lost	E-101 duty lost; the jacket alone cannot hold the peak: temperature and pressure rise to the trip	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % (no effect when the water is blocked) - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop - XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S5 L2 10 High
N8.1	Flow NO : Short-stop does not enter on demand	- XV-104 fails to open - D-105 empty - line plugged	Runaway continues to relief	- Operator ESD HS-001 - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C) - XV-104 proof test	S5 L1 5 Medium

HAZOP recommendation

LOPA for the runaway scenarios that rely on TSHH-101, PSHH-101, XS-101, FSLL-110 and the short-stop; set the SIL target and the proof-test interval of XV-104 and the logic solver.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N8 · Short-stop, emergency vent and relief	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.1	Flow NO : Short-stop does not enter on demand	• XV-104 fails to open • D-105 empty • line plugged	Runaway continues to relief	• Operator ESD HS-001 • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C) • XV-104 proof test	S5 L1 5 Medium
N8.2	Flow PART OF : Only part of the short-stop enters	D-105 pressure low (nitrogen supply valve closed, pot leaking)	Reaction slowed but not stopped; temperature may recover and rise again	• Operator can inject again from a refilled pot • PSV-100	S5 L1 5 Medium

HAZOP recommendation

Add a pressure transmitter with low-pressure alarm on the short-stop pot D-105 and make it an initiation permissive.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	N8 · Short-stop, emergency vent and relief	Action party	Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.6	Pressure MORE : Relief case reached	- Runaway not stopped by the short-stop - external fire	PSV-100 lifts: two-phase VCM/slurry to D-103; release if D-103 overfills	- PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C) - D-103 catch tank	S5 L1 5 Medium
N8.7	Flow NO : Relief path blocked	Polymer plugs the PSV-100 inlet or outlet	Vessel overpressure beyond design: rupture	- PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - inspection of the relief line	S5 L1 5 Medium

HAZOP recommendation

Confirm PSV-100 / XV-108 emergency-relief sizing for the runaway case with two-phase flow (DIERS method) and the capacity of D-103 for one full relief.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N5/N6 · Jacket tempered-water loop with CW and LP-steam injection / Baffle cooling water FIC-120	Action party	Process / Utilities
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.7	Utility NO : Cooling-water supply lost	CW pump trip or header failure (common to all four reactors)	Jacket, baffles and E-101 all lose cooling: runaway on every reacting batch	- FIC-120 LO 15 m³/h - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S5 L2 10 High
N5.8	Utility OTHER THAN : Cooling water warmer than 32 °C	- Cooling-tower performance - hot afternoon	Less capacity at the gel peak; TV-110B and E-101 at their limits	- TI-120 HI 35 °C (CW supply) - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 %	S3 L3 9 Medium
N6.1	Flow NO : No baffle water	- CW supply lost - baffle inlet blocked - controller fault closing FV-120	About 0.4 MW of cooling lost; TIC-100 shifts the load to the jacket and E-101	- FIC-120 LO 15 m³/h - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C	S3 L2 6 Medium

HAZOP recommendation

CW supply: low header-pressure alarm and CW supply flow on the reactor DCS; study CW failure as a common cause for all four reactors (stagger peak heat release, back-up CW pump on essential power).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N4 · R-101 during the reaction	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Temperature MORE : Reactor hotter than 57 °C	- Heat release above the removal capacity (gel-effect peak, warm CW, fouled wall, extra initiator) - loss of any cooling handle	K-value off grade, pressure rises with the VCM vapour pressure, runaway to relief	- TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L2 10 High
N4.3	Measurement OTHER THAN : Control element TE-100 reads low	Thermocouple drift or poor thermowell contact	TIC-100 holds the wrong temperature: the batch runs hot, K-value low, trip margin shrinks without a TIC-100 alarm	- TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S3 L3 9 Medium

HAZOP recommendation

Deviation alarm TE-100 vs TE-101 (1 °C) so a drifting control element is seen before the K-value or the trip margin is lost.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N4/N10 · R-101 during the reaction / Procedures	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.5	Measurement OTHER THAN : PT-100 frozen or wrong	Transmitter fault, plugged impulse line	PD-100 end point not seen: batch runs on in the pressure-drop phase (quality, cycle time) or is stopped early (VCM load on recovery)	- PI-101 (independent SIS transmitter) on the display - XI-100 conversion estimate	S2 L3 6 Medium
N9.4	Sequence BEFORE : Discharge before degassing is complete	- Interlock bypassed - PI-100 reads low	VCM flashes in D-102 and escapes at its vent: exposure and flammable atmosphere	XV-106 discharge blocked above 2.0 bar g and with the agitator stopped	S4 L2 8 Medium
N10.5	End point LATE : End point missed	- PD-100 not watched - PT-100 frozen	Batch runs on in the pressure-drop phase: quality drift, long cycle	- PD-100 END alarm - PI-101 comparison	S2 L3 6 Medium

HAZOP recommendation

Deviation alarm PT-100 vs PT-101 and a frozen-value check on PI-100, because the PD-100 end point depends on PT-100 alone.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N7/N8 · Reflux condenser E-101 and inert vent / Short-stop, emergency vent and relief	Action party	Process / Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.4	Pressure MORE : Pressure above the VCM vapour pressure	- Inerts (nitrogen passing, poor evacuation) collecting in E-101 - temperature high	E-101 duty falls, pressure and temperature rise; relief if uncontrolled	- PI-100 HI 11 bar g - TI-132 LO 46 °C (inerts at the E-101 top) - HIC-135 inert vent - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop	S4 L2 8 Medium
N7.4	Composition AS WELL AS : Inerts in E-101	- Nitrogen passing from the short-stop pot supply - poor evacuation - air in the charge	Inerts blanket the tubes: duty falls, pressure rises, TV-110B opens further	- TI-132 LO 46 °C (inerts at the E-101 top) - PI-100 HI 11 bar g - HIC-135 inert vent	S4 L2 8 Medium
N8.3	Flow AS WELL AS : Nitrogen into R-101 from D-105	Pressurising valve passing while the pot is connected	Inerts blanket E-101 (see N7.4)	- TI-132 LO 46 °C (inerts at the E-101 top) - PI-100 HI 11 bar g	S4 L2 8 Medium

HAZOP recommendation

Double block and bleed on the nitrogen supply to D-105 (open only to pressurise the pot); consider automatic inert venting on TI-132 low with a flow limit on HIC-135.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N7 · Reflux condenser E-101 and inert vent	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.2	Flow LESS : Less E-101 water than displayed	- FT-130 reads high - partly closed block valve	E-101 does less than the display says; TV-110B carries more; margin shrinks	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TI-131 HI 58 °C (E-101 CW return)	S3 L3 9 Medium

HAZOP recommendation

Calculated E-101 duty check ($FI-130 \times \Delta T$ against the heat balance) with a deviation alarm; include FT-130 in the calibration plan.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N7 · Reflux condenser E-101 and inert vent	Action party	Operations / Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Flow NO : No cooling water to E-101	- Block valve on the E-101 CW closed by mistake - CW supply lost	E-101 duty lost; the jacket alone cannot hold the peak: temperature and pressure rise to the trip	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % (no effect when the water is blocked) - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S5 L2 10 High
N7.2	Flow LESS : Less E-101 water than displayed	- FT-130 reads high - partly closed block valve	E-101 does less than the display says; TV-110B carries more; margin shrinks	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TI-131 HI 58 °C (E-101 CW return)	S3 L3 9 Medium

HAZOP recommendation

Car-seal open the E-101 CW block valves and add FI-130 LO alarm while E-101 is in service.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N4/N6/N7 · R-101 during the reaction / Baffle cooling water FIC-120 / Reflux condenser E-101 and inert vent	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.10	Composition AS WELL AS : Foam and polymer carried into E-101	Boil-up (E-101 water) raised too early in the batch	E-101 tubes and vapour line foul, condenser duty lost for later batches	- VPC-130 anti-foam limit (12 m³/h per % XI-100) - XA-133 foam alarm	S3 L3 9 Medium
N4.11	Heat transfer LESS : Polymer build-up on the reactor wall	- Wall scale from previous batches - poor cleaning	Jacket U falls; TV-110B opens further and the cooling margin at the peak shrinks	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C - wall cleaning between batches	S3 L3 9 Medium
N6.2	Flow LESS : Low baffle water	Strainer or baffle tubes scaled	Less baffle duty (small), hotter return	- FIC-120 opens FV-120 - TI-121	S2 L3 6 Medium
N6.5	Heat transfer LESS : Polymer on the baffles	Build-up from previous batches	Baffle duty falls over the campaign	- Cleaning between batches - TI-121 trend	S2 L3 6 Medium
N7.3	Flow MORE : Too much boil-up early in the batch	- FIC-130 in AUTO at a high set point - anti-foam limit bypassed	Foaming: PVC carried into the tubes and vapour line (fouling, blockage)	- VPC-130 anti-foam limit - XA-133 foam alarm	S3 L2 6 Medium
N7.5	Heat transfer LESS : E-101 fouled	- PVC deposits in the tubes (foam carry-over) - CW-side scale	Condenser duty falls; FIC-130 needs more water; less margin at the peak	- TI-131 HI 58 °C (E-101 CW return) - VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 %	S3 L3 9 Medium

HAZOP recommendation

Fouling programme: wall, baffle and E-101 cleaning interval set from the TV-110B and TI-131 trends; keep the VPC-130 anti-foam limit in service.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N1/N2/N4 · Demineralised water and PVA charge / VCM charge / R-101 during the reaction	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow MORE : Too much water charged	- FQI-101 preset entered too high - totaliser miscounts - XV-101 fails to close at the preset	Reactor overfilled: after VCM and heating the vapour space vanishes, liquid reaches E-101 and the vessel can go liquid-full → hydraulic overpressure	- LSHH-100 94 %, 5 s → charge valves and injection closed - LIC-100 HI 88 % / LO 74 % - water charge refused above 85 % level - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium
N2.2	Flow MORE : Too much VCM	- FQI-102 preset too high - totaliser error	Higher level and heat per volume; liquid-full on heating → overpressure; more heat at the gel peak than the cooling can remove	- LSHH-100 94 %, 5 s → charge valves and injection closed - LIC-100 HI 88 % / LO 74 % - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium
N4.8	Level MORE : Level above 83 %	- FV-103 passing - LT-100 radar reads low - LIC-100 set point high	Vapour space shrinks, foam and liquid into E-101 and the vapour line; liquid-full → hydraulic overpressure	- LIC-100 HI 88 % / LO 74 % - FIC-103 HI 8 m³/h - LSHH-100 94 %, 5 s → charge valves and injection closed - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium
N4.9	Level LESS : Level below 83 %	- Injection water stopped or isolated - LT-100 reads high	Wall above the liquid dries: polymer build-up, less jacket area (the injection exists to keep the wall wetted)	- LIC-100 HI 88 % / LO 74 % - WI-100 load cells	S2 L3 6 Medium

HAZOP recommendation

Confirm LSHH-100 uses its own level sensor (not LT-100) and add a check of LIC-100 against the load cells WI-100.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N4 · R-101 during the reaction	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.8	Level MORE : Level above 83 %	- FV-103 passing - LT-100 radar reads low - LIC-100 set point high	Vapour space shrinks, foam and liquid into E-101 and the vapour line; liquid-full → hydraulic overpressure	- LIC-100 HI 88 % / LO 74 % - FIC-103 HI 8 m³/h - LSHH-100 94 %, 5 s → charge valves and injection closed - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium

HAZOP recommendation

Second shut-off valve on the injection-water line closing on LSHH-100, and an alarm "FI-103 flow with FIC-103 output at zero".

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N5 · Jacket tempered-water loop with CW and LP-steam injection	Action party	Process / Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.4	Temperature MORE : Jacket inlet hotter than demanded	- TV-110B sticks - TIC-110 in MAN - LP steam passing TV-110A	Less heat removed: temperature rises until TIC-100 finds another handle or the trip acts	- TIC-100 → TIC-110 cascade, split range TV-110A steam (fail closed) / TV-110B CW (fail open) - TIC-100 HI 58 °C / LO 56 °C - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S4 L2 8 Medium
N5.6	Flow AS WELL AS : Steam enters the loop while the batch reacts	- TV-110A passing - split-range calibration error	Hidden heat load: more CW needed for the same temperature, margin at the peak lost	- TIC-110 corrects with TV-110B - TV-110B position on the DCS	S3 L3 9 Medium

HAZOP recommendation

Tight shut-off block valve on LP steam to SM-101, closed while the batch reacts, with position indication on the DCS.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N5 · Jacket tempered-water loop with CW and LP-steam injection	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.4	Temperature MORE : Jacket inlet hotter than demanded	- TV-110B sticks - TIC-110 in MAN - LP steam passing TV-110A	Less heat removed: temperature rises until TIC-100 finds another handle or the trip acts	- TIC-100 → TIC-110 cascade, split range TV-110A steam (fail closed) / TV-110B CW (fail open) - TIC-100 HI 58 °C / LO 56 °C - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S4 L2 8 Medium

HAZOP recommendation

Valve-position feedback on TV-110B with a deviation alarm against the TIC-110 output (stuck valve detection).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N1/N2/N10 · Demineralised water and PVA charge / VCM charge / Procedures	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.7	Composition AS WELL AS : Dissolved oxygen or air with the charge	- Water not de-aerated - air drawn in through the charge line	Induction period (oxygen inhibits), inerts that later blanket E-101, polyperoxide formation	- VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated) - TI-132 LO 46 °C (inerts at the E-101 top)	S2 L3 6 Medium
N2.4	Sequence BEFORE : VCM charged before the reactor is evacuated	- Evacuation cut short - XV-109 still open - permissive bypassed	Air and VCM in the vapour space: flammable mixture, polyperoxides, inerts that blanket E-101	- VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated) - VCM charge refused with XV-109 open	S4 L2 8 Medium
N10.1	Evacuation LESS : Reactor not fully evacuated	- Vacuum pump weak - XV-109 closed too early	Air left: induction period, inerts blanket E-101, oxygen in the VCM system	VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated)	S3 L2 6 Medium

HAZOP recommendation

Vacuum-hold test (pressure rise over 5 min) before the VCM charge; oxygen check of recovered VCM before re-use.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N2 · VCM charge	Action party	Process / Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Integrity OTHER THAN: VCM leak from the charge system	• Pump seal, flange or gasket failure • unloading hose • line hit during maintenance	Flashing VCM release: vapour cloud, flash fire or explosion; exposure to a carcinogen	• XV-102 fail closed (closed by any trip) • FV-102 fail closed • plant fire protection	S5 L2 10 High
N4.12	Integrity OTHER THAN: VCM leak from R-101	• Agitator seal failure • manway, bottom-valve or flange gasket failure	Flashing VCM release at 57 °C / 8.6 bar g: vapour cloud, flash fire or explosion; exposure to a carcinogen	• Agitator seal with barrier fluid (assumed) • XV-106 fail closed • ESD HS-001 closes the charges • plant fire protection	S5 L2 10 High

HAZOP recommendation

Remotely operated isolation valve at the TK-100 outlet and VCM gas detectors at P-103, the charge floor and the agitator seal, with automatic isolation of the charge.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N9 · Degassing to VCM recovery and discharge to D-102	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.3	Timing EARLY : Degassing a batch that is still reacting	XV-105 opened before the short-stop	Monomer vented from a live batch: evaporative cooling then foaming; recovery overloaded	Warning alarm "XV-105 OPEN" while reacting	S3 L2 6 Medium

HAZOP recommendation

Make short-stop (or end point reached) a hard permissive for opening XV-105; today the page only gives a warning alarm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	N9 · Degassing to VCM recovery and discharge to D-102	Action party	Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.4	Sequence BEFORE : Discharge before degassing is complete	- Interlock bypassed - PI-100 reads low	VCM flashes in D-102 and escapes at its vent: exposure and flammable atmosphere	XV-106 discharge blocked above 2.0 bar g and with the agitator stopped	S4 L2 8 Medium
N9.5	Flow OTHER THAN : Bottom valve opened on a reacting batch	- Wrong reactor selected in the field - XV-106 opened by mistake	Reacting slurry at 8.6 bar g into D-102: massive VCM flash and release	- XV-106 discharge blocked above 2.0 bar g and with the agitator stopped - XV-106 fail closed	S5 L1 5 Medium
N9.7	Level MORE : D-102 overfilled	- Previous batch not transferred - P-102 stopped	Slurry overflow at D-102, VCM release from the slurry	D-102 level indication (local)	S3 L2 6 Medium

HAZOP recommendation

Route the D-102 vent to VCM recovery, add D-102 high-level alarm and confirm D-102 holds one full batch plus margin.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N10 · Procedures	Action party	Operations / HSE
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.9	Sequence AFTER : N ₂ pad left open after R-101 is empty	HV-141 not closed at the end	Nitrogen fills the next batch (inerts in E-101); N ₂ in an opened reactor (asphyxiation)	End-of-batch procedure step (HV-141 closed)	S3 L2 6 Medium
N10.7	Opening AFTER : Reactor opened before VCM is removed	- Degassing or steam-out incomplete - VCM trapped in polymer build-up	Operators exposed to VCM, a carcinogen with a very low occupational limit; flammable atmosphere at the manway	- Permit to work with gas test - respiratory protection	S4 L3 12 High
N10.8	Opening OTHER THAN : Man-entry for cleaning	- Scraping build-up by hand - inspection	Confined space: VCM, nitrogen (asphyxiation), agitator movement, fall	- Confined-space permit - positive isolation and lock-out of M-101 - standby person	S4 L2 8 Medium

HAZOP recommendation

Reactor opening: degas to recovery, steam-strip and evacuate until a portable VCM measurement at the manway is below the occupational limit; closed-reactor high-pressure water cleaning so that man-entry is the exception; confined-space permit with positive isolation.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N1/N2/N10 · Demineralised water and PVA charge / VCM charge / Procedures	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow MORE : Too much water charged	- FQI-101 preset entered too high - totaliser miscounts - XV-101 fails to close at the preset	Reactor overfilled: after VCM and heating the vapour space vanishes, liquid reaches E-101 and the vessel can go liquid-full → hydraulic overpressure	- LSHH-100 94 %, 5 s → charge valves and injection closed - LIC-100 HI 88 % / LO 74 % - water charge refused above 85 % level - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium
N1.3	Flow LESS : Less water than the recipe	- FQI-101 preset low - header pressure low and the charge stopped early	Higher solids and viscosity, lower jacket heat transfer, hotter batch, more lumps	- VCM charge permissive (water ≥ 95 % of preset) - LIC-100 LO alarm - JI-100 HI alarm	S3 L2 6 Medium
N1.6	Composition PART OF : Only part of the PVA reaches the reactor	- HV-107 opened late or not at all - PVA pot not refilled - wrong concentration	Unstable suspension: droplets coalesce, lumps grow, agitator power and wall fouling rise, jacket U falls; bottom valve may block at discharge	- Start-up procedure step (PVA ≥ 95 % of recipe) - JI-100 LO 150 kW / HI 420 kW	S3 L3 9 Medium
N2.2	Flow MORE : Too much VCM	- FQI-102 preset too high - totaliser error	Higher level and heat per volume; liquid-full on heating → overpressure; more heat at the gel peak than the cooling can remove	- LSHH-100 94 %, 5 s → charge valves and injection closed - LIC-100 HI 88 % / LO 74 % - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium
N10.9	Recipe OTHER THAN : Wrong presets for the batch	Grade change, typing error in FQI presets or initiator amount	Over- or under-charge (see N1.2, N2.2, N3.1)	Operator check against the recipe sheet	S3 L2 6 Medium

HAZOP recommendation

Recipe presets (FQI-101, FQI-102, initiator) downloaded from a recipe system and checked by a second person before the charge.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N7/N9/N10 · Reflux condenser E-101 and inert vent / Degassing to VCM recovery and discharge to D-102 / Procedures	Action party	Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.6	Flow MORE : Inert vent left open	HIC-135 left at a high output after venting	VCM vented continuously to recovery: recovery load, less monomer in the batch	- FI-135 indication - operator	S2 L3 6 Medium
N7.7	Pressure MORE : Recovery back-pressure high	- Recovery compressor C-101 tripped - gasholder full	Inerts cannot be vented; degassing slows (see N9)	- PI-100 - recovery operator	S3 L2 6 Medium
N9.2	Flow LESS : Degassing stalls	- Recovery compressor tripped - gasholder full (back-pressure)	Pressure stays high; cycle time lost; temptation to discharge early	- Discharge interlock (PI-100 below 2.0 bar g) - PI-100	S3 L2 6 Medium
N9.8	Flow REVERSE : Recovery gas back into R-101	XV-105 left open with R-101 below the recovery pressure	Gas from other reactors enters R-101 (inerts, VCM in an opened reactor)	XV-105 closed by procedure below 2 bar g	S2 L2 4 Low
N10.2	Evacuation OTHER THAN : Vacuum pump run on a reactor containing VCM	XV-109 opened with VCM still in R-101 (after degassing, cleaning)	VCM through VP-101 to the vent: exposure and flammable mixture at the vent	XV-109 refused above 1.0 bar g	S4 L2 8 Medium

HAZOP recommendation

Show VCM recovery status (compressor C-101 running, gasholder D-106 level) on the reactor DCS; keep vacuum-pump exhaust away from the recovery system (vent treatment only).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	All · whole unit	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow MORE : More initiator than the recipe	- Weighing error - pot flushed twice (second dose) - wrong concentration	Heat release beyond the cooling capacity: temperature and pressure rise, runaway, relief of VCM	- VPC-130 moves E-101 water (FIC-130, FV-130 fail open) to hold TV-110B near 50 % - TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101) - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L2 10 High
N3.5	Timing LATE : Initiator dosed while the jacket is still hot from heat-up	- Steam heat-up not stopped in time - TIC-110 left on high jacket temperature	Temperature overshoot in the first hour as the fast initiator starts: off-grade K-value, approach to the trip	- TIC-110 HI 75 °C alarm while the initiator is in (first hour) - TIC-100 HI 58 °C / LO 56 °C - TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open	S3 L2 6 Medium
N4.7	Agitation LESS : Agitation lost with the motor running	- Coupling or shaft failure - impeller loose	As above but XS-101 does not see it (motor shows RUN); lumps grow; heat transfer falls	- JL-100 LO 150 kW / HI 420 kW - TIC-100 via more jacket CW and E-101 water	S5 L2 10 High
N5.9	Control OTHER THAN : Temperature cascade broken	- TIC-110 left in AUTO/MAN after heat-up - TIC-100 in MAN	Reactor temperature no longer controlled; drift with the heat release	- TIC-100 HI 58 °C / LO 56 °C - TI-101 HI 60 °C (independent TE-101)	S3 L3 9 Medium
N7.9	Control OTHER THAN : E-101 not on valve-position control	VPC-130 in MAN or FIC-130 not in cascade	E-101 fixed: the jacket takes all swings and TV-110B saturates at the peak	- TIC-100 HI 58 °C / LO 56 °C - TV-110B on the DCS	S3 L3 9 Medium
N10.4	Initiation EARLY : Initiation while the batch is still being heated hard	Steam heat-up not stopped before dosing	Overshoot above 58 °C in the first hour, off-grade K-value, approach to the trip	- TIC-110 HI 75 °C alarm while the initiator is in (first hour) - TIC-100 HI 58 °C / LO 56 °C	S4 L2 8 Medium

HAZOP recommendation

Train operators on the OTS for loss of agitation, loss of cooling, initiator overcharge and hot initiation, including when to short-stop by hand.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R28
Node	N8 · Short-stop, emergency vent and relief	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.4	Sequence BEFORE : VCM charged before the reactor is evacuated	- Evacuation cut short - XV-109 still open - permissive bypassed	Air and VCM in the vapour space: flammable mixture, polyperoxides, inerts that blanket E-101	- VCM charge permissive: PI-100 below water vapour pressure + 0.15 bar a (reactor evacuated) - VCM charge refused with XV-109 open	S4 L2 8 Medium
N8.9	Safety system OTHER THAN : SIS initiator bypassed	Bypass left in after testing or during a transmitter fault	The runaway protection is missing when needed	Administrative control only	S5 L2 10 High

HAZOP recommendation

Trip-bypass management: any SIS bypass needs a permit, a time limit and MOC; bypass status shown on the DCS.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R29

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R29
Node	N1 · Demineralised water and PVA charge	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.6	Composition PART OF : Only part of the PVA reaches the reactor	• HV-107 opened late or not at all • PVA pot not refilled • wrong concentration	Unstable suspension: droplets coalesce, lumps grow, agitator power and wall fouling rise, jacket U falls; bottom valve may block at discharge	• Start-up procedure step (PVA $\geq$ 95 % of recipe) • JI-100 LO 150 kW / HI 420 kW	S3 L3 9 Medium

HAZOP recommendation

Measure the PVA charge (flow totaliser or TK-107 load cell) and make it a VCM-charge permissive.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R30

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R30
Node	N1/N2/N3 · Demineralised water and PVA charge / VCM charge / Initiator charge	Action party	Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.9	Flow REVERSE : Reactor contents back into the water header	XV-101 and FV-101 passing while R-101 runs at 8.6 bar g, above the header pressure	VCM enters the demineralised-water system and is released at other users (exposure, fire)	• XV-101 and FV-101 both fail closed and closed by the sequence/SIS • header pressure alarm (utility)	S4 L2 8 Medium
N2.7	Flow REVERSE : Reactor contents back to TK-100	P-103 stopped with XV-102/FV-102 open while R-101 is above the storage pressure	Slurry and initiator into the VCM store: polymerisation and fouling in storage, overpressure of the store	• XV-102 closes at the preset (FQI-102) • XV-102 and FV-102 fail closed	S4 L2 8 Medium
N3.8	Flow REVERSE : Reactor contents into D-104	XV-103 left open after dosing with P-104 stopped (R-101 at 8.6 bar g)	VCM and slurry contaminate the peroxide pot → decomposition, pot overpressure	• XV-103 fail closed and closed after the 60-s dose • dosing pump positive displacement	S4 L1 4 Low

HAZOP recommendation

Check valves plus the FC charge valves on every line into R-101 (water, VCM, initiator); confirm back-flow protection because R-101 runs above the water header and the storage pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R31

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R31
Node	N2 · VCM charge	Action party	Mechanical
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.5	Pressure MORE : P-103 runs against a closed valve	XV-102 or FV-102 closes (preset, LSHH-100, trip) with P-103 still running	Pump overheats; seal failure → VCM leak at the pump (fire, exposure)	- Operator stops P-103 after the charge - pump seal	S4 L2 8 Medium
N2.6	Integrity OTHER THAN : VCM leak from the charge system	- Pump seal, flange or gasket failure - unloading hose - line hit during maintenance	Flashing VCM release: vapour cloud, flash fire or explosion; exposure to a carcinogen	- XV-102 fail closed (closed by any trip) - FV-102 fail closed - plant fire protection	S5 L2 10 High

HAZOP recommendation

Minimum-flow recycle to TK-100 on P-103 and a double mechanical seal with barrier-pressure alarm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R32

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R32
Node	N2 · VCM charge	Action party	Laboratory
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.8	Composition AS WELL AS : Impurities in the VCM (oxygen, acetylene, HCl, water)	Recovered VCM returned to storage out of specification	Induction period, slower batch, off-grade K-value, polyperoxide hazard with oxygen	Laboratory analysis of fresh VCM	S2 L3 6 Medium

HAZOP recommendation

Specification check of recovered VCM before re-use (oxygen, acetylene, HCl, water).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R33

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R33
Node	N3 · Initiator charge	Action party	Laboratory / Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Flow LESS : Less initiator activity	• Aged or warm-stored lot • under-weighed dose	Slow batch, long cycle, more VCM left at the end (operability, recovery load)	• XI-100 against the reference batch • heat-balance display	S2 L3 6 Medium
N3.6	Temperature MORE : Initiator pot or store warms up	• Refrigeration failure • drum left at ambient • heat from the reactor bay	Self-accelerating decomposition of the peroxide: pot overpressure, fire, toxic fumes	• Chilled pot and store • small pot inventory • temperature indication at the store	S4 L2 8 Medium
N3.7	Composition OTHER THAN : Wrong initiator	Wrong drum or grade picked (faster peroxide)	Very fast start, heat peak early and above the cooling capacity	• TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open • PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop • PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S5 L1 5 Medium

HAZOP recommendation

Initiator lot control: assay certificate, first-in-first-out, store temperature records; reject drums that have been warm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R34

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R34
Node	N6/N7 · Baffle cooling water FIC-120 / Reflux condenser E-101 and inert vent	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.4	Integrity OTHER THAN: Baffle tube leak	Corrosion, fatigue	R-101 (8.6 bar g) is above the CW pressure: VCM enters the CW return and is released at the cooling tower (exposure, fire)	None specific	S4 L2 8 Medium
N7.8	Integrity OTHER THAN: E-101 tube leak	Corrosion or vibration	VCM (reactor side at higher pressure) into the CW return and the cooling tower	None specific	S4 L2 8 Medium

HAZOP recommendation

VCM detection on the CW returns from the baffles, jacket overflow and E-101 (tube-leak detection) before the cooling tower.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R35

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R35
Node	N8 · Short-stop, emergency vent and relief	Action party	Mechanical
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.7	Flow NO : Relief path blocked	Polymer plugs the PSV-100 inlet or outlet	Vessel overpressure beyond design: rupture	- PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - inspection of the relief line	S5 L1 5 Medium

HAZOP recommendation

Rupture disc with pressure-monitored interspace under PSV-100 to stop polymer plugging the relief path; inspect the relief line by batch count.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R36

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R36
Node	N8 · Short-stop, emergency vent and relief	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.5	Flow MORE : Short-stop injected without need	- Spurious trip - operator error	Batch killed at low conversion: off-grade, VCM to recover (operability)	- Independent trip transmitters - trip first-out display	S2 L3 6 Medium

HAZOP recommendation

Review the spurious-trip rate of the SIS initiators (e.g. 2oo3 voting on temperature and pressure) together with the SIL target from R06.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R37

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R37
Node	N9 · Degassing to VCM recovery and discharge to D-102	Action party	Process
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.1	Flow MORE : Degassing too fast	HIC-105 opened fully at once	Foaming: slurry and PVC carried into the recovery line, gasholder and compressor fouled	- Operator throttles HIC-105 - FI-105 indication	S3 L3 9 Medium

HAZOP recommendation

Degassing rate limit (ramp on HIC-105) and a knock-out pot ahead of the gasholder to catch foam and slurry.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R38

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R38
Node	N10 · Procedures	Action party	Operations
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.4	Timing LATE : Short-stop injected when the agitator has already stopped	- Power failure stops M-101 - short-stop by hand after the stop	Short-stop does not mix (minutes instead of seconds): local reaction goes on in the separated mass	- XS-101 agitator stopped while reacting, 15 s → short-stop (injects at once) - emergency vent XV-108	S5 L2 10 High
N10.10	Restart OTHER THAN : Agitator restarted on a stopped, still-reacting batch	Motor reset and restarted without confirming the short-stop	Sudden mixing of hot, unreacted monomer: heat surge beyond the cooling, pressure rise	- TSHH-101 on TE-101 (independent of TE-100) at 63 °C, 10 s → short-stop XV-104, charges closed, TV-110B/FV-120/FV-130 forced open - PSHH-101 on PT-101 (independent of PT-100) at 12.5 bar g, 5 s → short-stop - PSV-100 set 15 bar g to dump tank D-103 (vessel design 15.2 bar g / 93 °C)	S4 L2 8 Medium

HAZOP recommendation

Written rule for restarting the agitator after it has stopped on a batch that is still reacting (only after the short-stop has been confirmed, with full cooling), practised on the OTS.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R39

Project	PVC Suspension Reactor Ver 1 · R-101 Operator Training Simulator	Document	PVC-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R39
Node	N8 · Short-stop, emergency vent and relief	Action party	Instrument
Drawings	PVC-PID-001 (R-101, charges, condenser, vents) and PVC-PID-002 (heat removal); simulator P&ID tab	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.8	Utility NO : Instrument air lost	Air compressor trip, header break	XV-104 and XV-108 fail closed and cannot open on demand; cooling valves fail open (safe side)	• TV-110B, FV-120, FV-130 fail open (full cooling) • charge valves fail closed	S3 L2 6 Medium

HAZOP recommendation

Dedicated instrument-air receiver (with check valve) for XV-104 and XV-108 so the short-stop and emergency vent can still open after an air failure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			